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# MMR(IRIS1,IRIS2) Manual



|                                                                                   |                            |          |                  |
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## Danger

Due to the Main Power ON, High voltage is still charged in the inside.

Be sure to the main power OFF when opening the door of a controller cabinet.

Otherwise, there may be a fatal damage by an electric shock.



## Warning

If you do not keep the procedures specified on this manual, there may be controller damage or an injury of workers.

1. Prior to Main Power ON, be sure that the 3 phase of Main Power is connected to R, S, T terminal correctly.
2. Keep the shield treatment and the process specified in this Manual, and do not work in the lift and/or on the car top without confirming the lift is moving stable from the machine room.

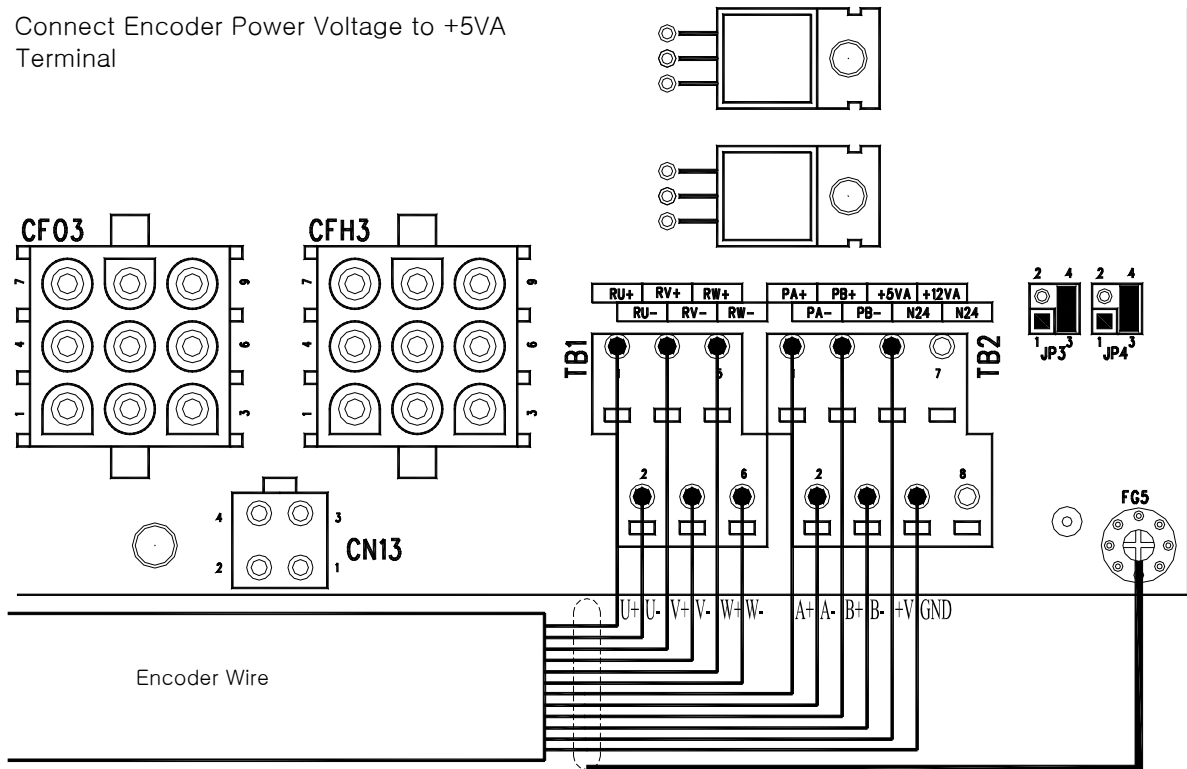


## Electrical shock

If you do not keep this mark indicates that the procedures there may happen an electrical shock.

### \* Encoder Wiring

Connect Encoder Power Voltage to +5VA Terminal



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## 1. 1. User's Manual

### 1. MMR MANUAL

#### 1. The outline of product

This is a user's manual for MMR controller.

MMR is, as an Elevator Controller, combining micom control PCB and Inverter PCB into the one PCB, compact and supply enhanced performance.

#### 1.1 Attention before installation

- Do not wrongly connect 3 phase power Input Line (R,S,T) and Output Line (U,V,W). If wrongly connected, inverter may be damaged.
- Check the U, V, W wires to motor whether they are not exchanged each other.
- Use shielded-wires and/or do shielding for U, V, W to motor, encoder pulse input wires and regenerative resistor wires.
- Never touch the output terminals directly with your hands if charge lamp in the control board is ON. Especially do not touch control power board. There may be potentially electrical shock hazard
- Storage suggestions are:
  - Store within an ambient temperature range of -10°C to 45 °C
  - Store within a relative humidity range of 0% to 90%, non-condensing
  - Do not store the drive in places where it could be exposed to a corrosive and combustible atmosphere.  
Do not store the drive on a shelf or on an unstable surface
  - Store in a clean, dry position  
Do not store in direct sunlight  
Install in a position without excessive vibration or impulse
- Do not changeover the jumper pin, which is pre-set in factory, on control panel. Otherwise the inverter shall not be operated.

|                                                                                   |                            |          |                  |
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## 1.2 Characteristics

This controller provides fully digital vector control based on advanced control logic. Auto-tuning function is included for easy commissioning.

### 1. Direct landing

Direct door opening is possible after landing, because Micom controls actual speed of motor through real time monitoring.

### 2. Creepless operation

Combining control PCB & Inverter PCB, creepless operation is possible.

### 3. Convenient setting

Using operator, OPP-2000, Operation condition can be easily adjusted.

### 4. 100 % digital

Adopting DSP (Digital Signal Processor) chip, as current control as well as speed control are done by high-speed software, there are no time-variances of performances and also shows excellent realization.

### 5. Abundant monitoring functions

With command speed, actual speed, maximum torque, actual torque, output voltage, DC bus voltage, DIO(Digital Input/Output) information, as all operation condition can be seen through OPP-2000, and total 10 past alarms storage, it is easy to operate the system and do maintenance.

### 6. Characterized function for Elevator

Easily setting S-curve acceleration & deceleration makes system simplify.

### 7. Low Audible noise

Since IPM(Intelligent Power Module) power electronics element, is adopted, for all model and also using high speed switching the motor does not make audible noise.

### 8. Lock

Lock Function detect automatically Encoder character.

This Function should be done after lift stops.



|                                                                                   |                            |          |                  |
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## 9. Serial Communication

Signal transmission is done by Serial Communication and wires consist of 6 wires including power line.

## 10. Record for Failure Detection

130 Kinds of Error Detection and maximum 640 Failure Record is possible.

## 11.Password Function

Duplicated Password Function authorization to use controller.

## 12.Group control

Maximum 4 car Group control is possible.

## 2. Check-up when the Goods arrived.

**Check MMR when it is arrived as follows.**

**As a result, In case of any discrepancy, contact us.**

- 1) Check the things compared with purchase order.
- 2) Check if there is any damage during transportations.

Please follow the instructions bellows.



- 1) Read the operational manual before operation.
- 2) Set the system according to the operational manual.
- 3) Do the test operation after other safe conditions sufficiently.
- 4) As CMOS is used on the control board, when it is touched by hand directly, the product could be damaged by the static electricity. So, please be careful of handling.

|                                                                                   |                            |          |                  |
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- 5) Connect the power after confirming the input power voltage.
- 6) Make sure that not to change the wires when applying an electric current,  
be sure to change the wires after turning off the power.
- 7) Boards should be changed after the power is turned off.
- 8) Pay attention to signal check during operation.
- 9) Be sure to earth the earth terminal.

**When carrying the control panel, pay attention to the details as below.**

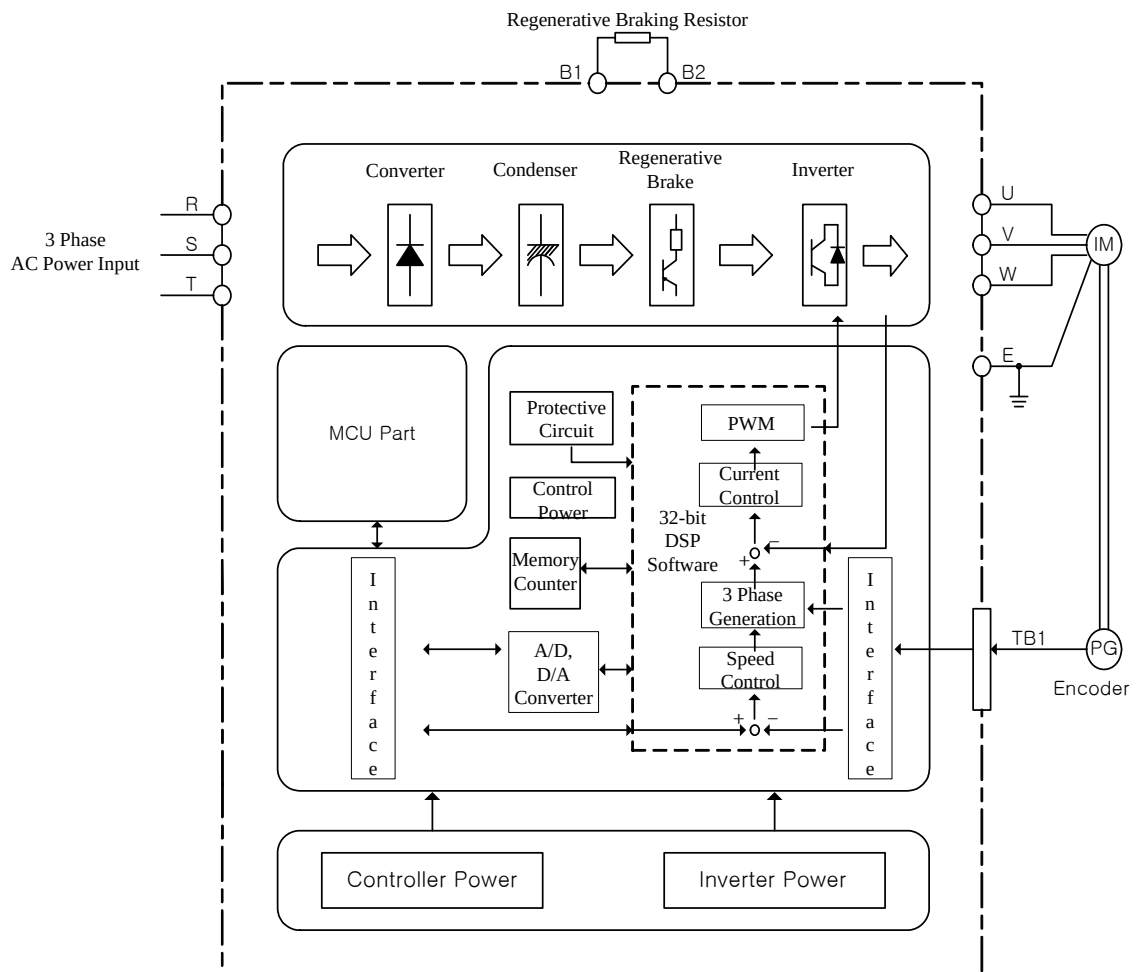
- 1) Be careful not to be damaged when carrying it.
- 2) Hold whole control cabinet, not door.

|                                                                                   |                            |          |                  |
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### 3. Configuration and Specification

#### 3.1 CONFIGURATION OF CONTROLLER

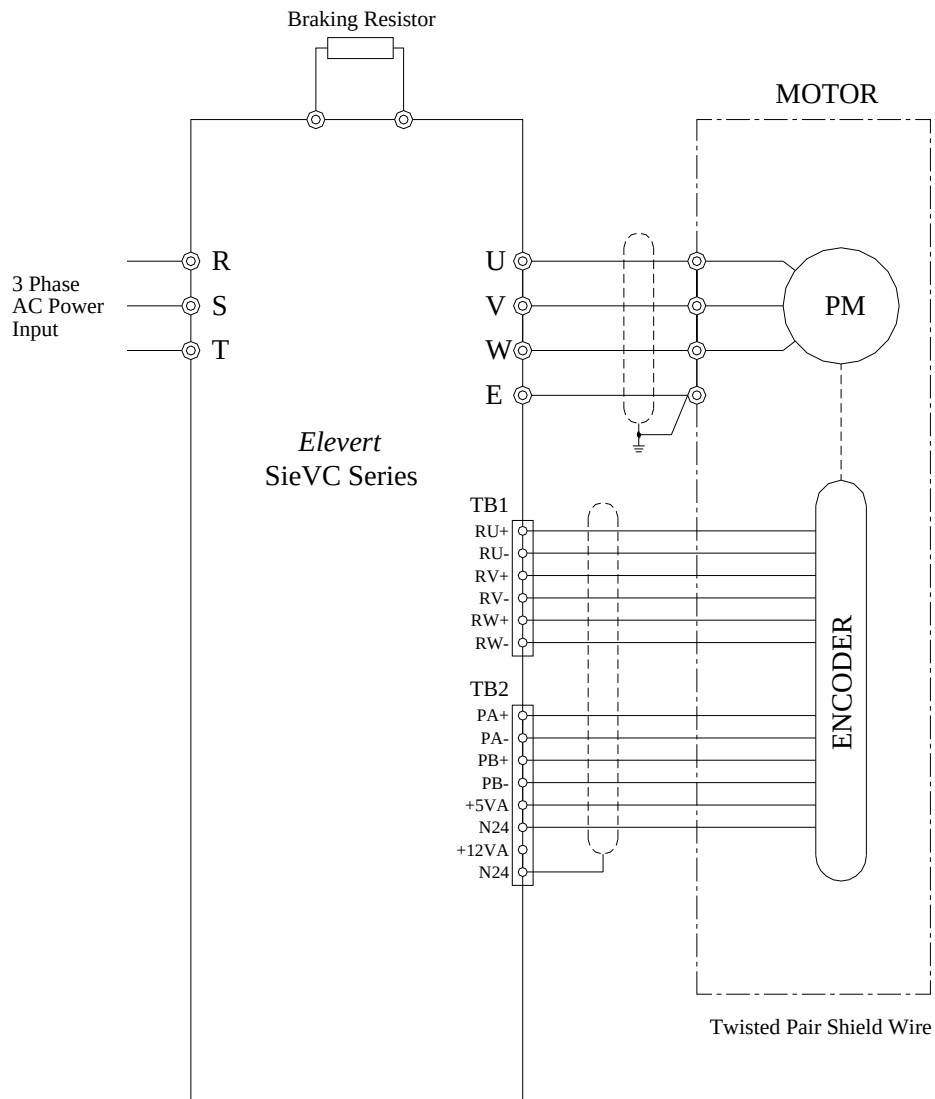
Configuration of controller is as shown below. Refer to Fig.-2 for wiring.



**Fig.-1. Configuration of MMR**

|                                                                                   |                            |          |                  |
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Fig.-2. Basic Wiring



|                                                                                   |                            |          |                  |
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### 3.2 DESCRIPTION OF CONTROLLER MODEL AND STANDARD SPECIFICATIONS

Table-1 shows controller models and Table-2 shows standard specifications.

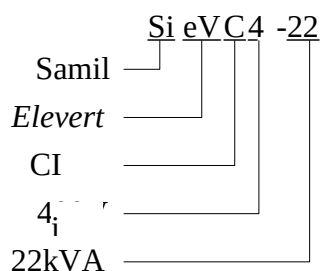


Table-1 Type Symbol

Table-2. *Elevert-Ci*® standard specification

| 200V Series     |                          | eVC2-5.5                                                 | eVC2-7.5 | eVC2-11 | eVC2-15 | eVC2-22 |
|-----------------|--------------------------|----------------------------------------------------------|----------|---------|---------|---------|
|                 | Capacity(kVA)            | 10                                                       | 13       | 17      | 25      | 32      |
|                 | Rated Current(A)         | 27                                                       | 37.3     | 49.5    | 73      | 93      |
|                 | Overload Capacity        | 200% of rated current for 1 second                       |          |         |         |         |
|                 | Voltage • Frequency      | 3-Phase 200V ~ 230V, 50 / 60 Hz                          |          |         |         |         |
|                 | Frequency Tolerance      | ± 5%                                                     |          |         |         |         |
| 400V Series     |                          | eVC4-5.5                                                 | eVC4-7.5 | eVC4-11 | eVC4-15 | eVC4-22 |
|                 | Capacity(kVA)            | 10                                                       | 10       | 17      | 24      | 33      |
|                 | Rated Current(A)         | 13                                                       | 17       | 24.7    | 34      | 47      |
|                 | Overload Capacity        | 200% of rated current for 1 second                       |          |         |         |         |
|                 | Voltage • Frequency      | 3-Phase 380V ~ 430V, 50 / 60 Hz                          |          |         |         |         |
|                 | Frequency Tolerance      | ± 5%                                                     |          |         |         |         |
| Characteristics | Control system           | High frequency Space Vector PWM (Pulse Width Modulation) |          |         |         |         |
|                 | Speed regulation         | ±0.01%                                                   |          |         |         |         |
|                 | Max. torque              | 200% of rated torque                                     |          |         |         |         |
|                 | Signal for speed setting | 3-Bit Binary                                             |          |         |         |         |

|                 |                       |                                                                         |
|-----------------|-----------------------|-------------------------------------------------------------------------|
|                 | Accel./Decel. Time    | 0.01second(Independently settable for accel., decel., and creeping)     |
|                 | S-curve setting       | Independently settable for accel. start/stop, decel. start/stop         |
| Function        | Motor Protection      | Protected by electronic thermal function                                |
|                 | Over-current          | 200% of rated output current                                            |
|                 | Over-voltage          | Halt when DC bus voltage is over 400V (800V for 400V series)            |
|                 | Under-voltage         | Halt when DC bus voltage is below 190V(380V for 400V series)            |
|                 | Over-speed            | Halt when speed is over $\pm 110\%$ of synchronous rpm                  |
|                 | Speed error Over      | Halt when speed error is over $\pm 20\%$ of synchronous rpm             |
|                 | Overheat of Heat-sink | By thermistor                                                           |
| Environ<br>ment | Installation Location | Keep from corrosive gasses, liquid and dust                             |
|                 | Ambient Humidity      | Below 90%(non-condensing)                                               |
|                 | Storage Temperature   | -20 ~ +60 °C                                                            |
|                 | Ambient Temperature   | -10 ~ +45 °C                                                            |
|                 | Altitude              | 1000m or lower                                                          |
|                 | Vibration             | Less than 1G at vibration of 10 ~ 20Hz,<br>Less than 0.2G at 20 ~ 50 Hz |

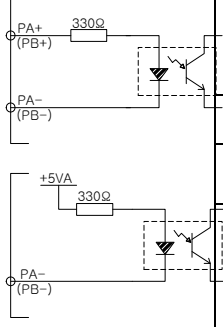
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## 4. EXPLANATION OF FUNCTIONS

### 4.1 TERMINAL SIGNAL SPECIFICATIONS

The controller is connected by terminal block TB, TB1 and the function of each signal is described as below.

**Table –3 Signal Specifications**

| Terminal | Symbol   | Signal specification                                                                | Function                   |
|----------|----------|-------------------------------------------------------------------------------------|----------------------------|
| TB       | R        | 3-phase power input terminal                                                        |                            |
|          | S        |                                                                                     |                            |
|          | T        |                                                                                     |                            |
|          | E        | Earth(Ground) terminal                                                              |                            |
|          | U        | 3-phase output terminal to motor                                                    |                            |
|          | V        |                                                                                     |                            |
|          | W        |                                                                                     |                            |
|          | E        | Earth(Ground) terminal                                                              |                            |
|          | B1<br>B2 | Terminals for external regenerative braking resistor                                |                            |
| TB2      | PA +     |  | A+ phase Encoder input     |
|          | PA -     |                                                                                     | A- phase Encoder input     |
|          | PB +     |                                                                                     | B+ phase Encoder input     |
|          | PB -     |                                                                                     | B- phase Encoder input     |
|          | +5VA     | -                                                                                   | Encoder power supply, +5V  |
|          | N24      | -                                                                                   | Ground for Encoder Power   |
|          | +12VA    | -                                                                                   | Encoder power supply, +12V |
|          | N24      | -                                                                                   | Ground for Encoder Power   |

TB2: Connector for the connection of encoder signals

|                                                                                   |                            |          |                  |
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## 5. Installation.

### 5-1. Caution when installing.



- 1) When carrying the MMR, please hold the whole controller cabin.
- 2) If you take the terminal or front cover, it can cause damage.
- 3) Avoid the installation places like bellows.
  - ⇒ The place where a direct ray of light rays.
  - ⇒ The place where surrounding temperature is below  $-10^{\circ}\text{C}$  and over  $45^{\circ}\text{C}$ .
  - ⇒ The place where facing humidity is over 90%.
  - ⇒ The place where there is possibility of appearing of dew by severe temperature change.
  - ⇒ The place where combustible gas or rusty gas exist.
  - ⇒ The place where there is possibility of being touched by rain or water.
  - ⇒ The place where there is possibility of impacting or vibration to the control panel.
  - ⇒ The place affected by electronic noise.
- 3) Do not wrongly connect 3phases power input lines(R, S ,T) and output lines(U, V, W)to the motor.  
Otherwise, it can cause severe damage.
- 5) The lines to the motor(U, V, W), encoder pulse input line and restoration resistance connection line, should be covered by shield.
- 6) As there is a high-voltage part in the Inverter, when charging lamp is on, do not touch a board inside the inverter. Otherwise, it can cause a electric shock.
- 7) Output line U, V, W to the motor should not be changed.
- 8) Make it sure of connection of connector. Most of all problems is caused by poor, wrong connection. Connect according to the drawing.
- 9) Earth should be 3 splices.



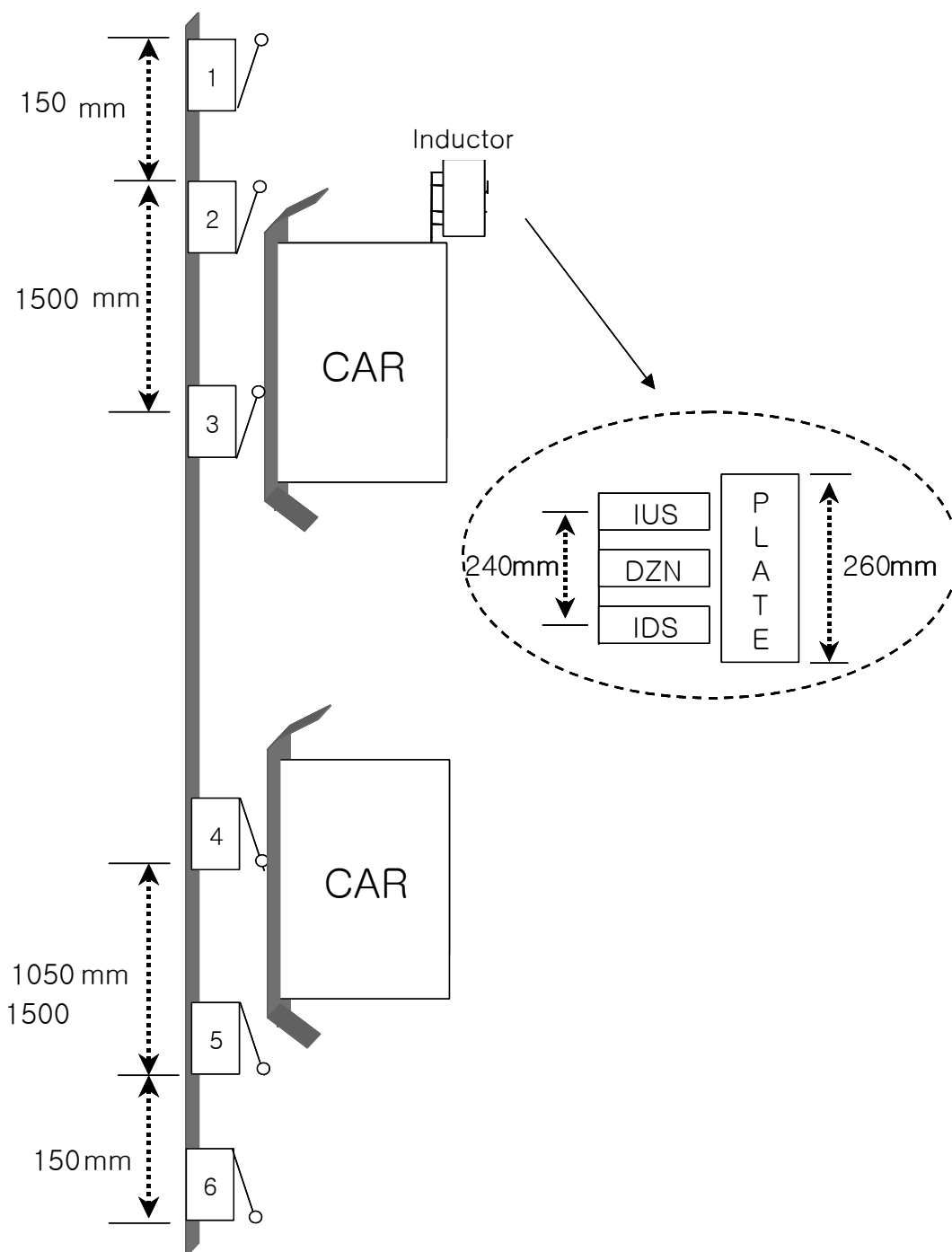
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### 5-2. Attention to wiring



- 1) Connect the wires exactly according to the drawing.
- 2) Wrong connection can cause the damage of the product and give rise to an accident. So, please connect wires exactly.
- 3) Signal lines should be placed as far as possible from power lines.  
(R, S, T, U, V, W).
- 4) If signal lines and power lines are in the short distance, it can cause a strange effect because of noise to the signal lines.
- 5) The wires you use should be standard as drawing shows.
- 6) If the wires connected wrongly, it can cause a fatal damage to the product and wrong action.
- 7) Earth line should be connected to the earth of control panel firmly.
- 8) Confirm whether R,S,T and U.V.W lines are connected in proper.

### 5-3. Installation of Limit switch and Plate (In case of 60m / min)



- 1. UFL: Up Final Limit
- 3. USL: Up Slow Limit
- 5. DL: Down Limit

- 2. UL: Up Limit
- 4. DSL: Down Slow Limit
- 6. DFL: Down Final Limit

|                                                                                   |                            |          |                  |
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#### 5-4. Hoistway Installation

- 1) Limit switch and plate should be installed referring of 5-3 in the hoist way.
- 2) The length of plate on each floor should be 200mm.
- 3) Both IUS and IDS should be installed and DZN is optional.
- 4) If DZN(Door Zone Switch) is not installed, Door Zone section becomes in case of any one of IUS or IDS. Accordingly, when you use DZN, door zone section is +/- 130mm and if not, +/- 250mm.(In case the length Plate is 260mm). When you use DZN switch, select [YES] on the OPP-2000 [OPT] Menu.
- 5) Install UL (Up Limit Switch) and DL (Down Limit Switch) don't work when a car is on the level. If not, a car doesn't start in automatic operation.
- 6) USL(Up Slow Limit) and DSL(Down Slow Limit) should be turned on when a car is on the highest or lowest level. If not, it causes "Floor Error".
- 7) All the wires should be in order in the hoistway, so that they don't get tangled when a car moves.

#### 5-5. Installation of indicator at each floor



- 1) All indicators used on MMR is serial communication method and have function of sending and receiving.
- 2) Set the floor setting on the board referring to Indicator Board Manual. When it's done, be sure not to floor overset. In case setting of the dip switch is changed, the power should be reset.  
6 Pin (Communication Line) to the harness is exceptional. Other connectors on the board should be pulled out after 6 Pin (communication Line) to the harness is off.
- 3) All Dip switch on the board inside a car should be uplift and on. (For example, COP, voice synthesizer connection board etc.)

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## 5-6. Installation of Door Driver



- 1) The installation of Door operation should be done according to the system. (Refer to drawing)
- 2) As DCL(Door Close Limit) and DOL(Door Open Limit) are used not only door operation device but also control micom board, they should be connected according to electronic drawing. But, the signal of DOL, DCL are not used in manual operation.

|                                                                                   |                            |          |                  |
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## 6. Starting Manual

### 6-1. Before operation

#### 6-1-1. Basic Installation



- 1) Check R,S,T and U,V,W are connected in proper.
- 2) Check Encoder is connected exactly.
- 3) Door should be open and close in proper.
- 4) Limit switch should be installed.(UL,USL,DL,DSL etc)
- 5) Set the Limit contact and driving speed on the SMCB-3000Ci Board according to the system.

| SW1                                                                                                                                                                                                                                                                                                                                        | ①   | ②   | ③   | ④   | Explanation                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|----------------------------------------|
| <div> <div>ON</div> <div>OFF</div> <div> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> </div> <div> <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> </div> <div>① ② ③ ④</div> </div> | OFF | OFF | OFF | -   | 1 floor operation                      |
|                                                                                                                                                                                                                                                                                                                                            | OFF | OFF | ON  | -   | 2 floor operation(60m/mn)              |
|                                                                                                                                                                                                                                                                                                                                            | OFF | ON  | OFF | -   | 3 floor operation<br>(90~105m/min)     |
|                                                                                                                                                                                                                                                                                                                                            | OFF | ON  | ON  | -   | 4 floor operation<br>(120m/min)        |
|                                                                                                                                                                                                                                                                                                                                            | ON  | OFF | OFF | -   | 5 floor operation<br>(150m/min)        |
|                                                                                                                                                                                                                                                                                                                                            | -   | -   | -   | OFF | Limit switch contact is<br>"B" contact |
|                                                                                                                                                                                                                                                                                                                                            | -   | -   | -   | ON  | Limit switch contact is<br>"A" contact |

- 6) Select simplex or group on the SMCB-3000Ci Board.

| SW2                                                                                                                                                                                         | ①         | ②         | ③         | ④         | Explanation                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|-----------|---------------------------------------------|
| <div> <div>ON</div> <div>OFF</div> <div> <div>□</div> <div>□</div> <div>□</div> <div>□</div> </div> <div> <div>■</div> <div>■</div> <div>■</div> <div>■</div> </div> <div>①②③④</div> </div> | OFF       | -         | -         | -         | Simplex(SKA)                                |
|                                                                                                                                                                                             | <b>ON</b> | -         | -         | -         | Duplex,<br>Group(DSKA,N-PLEX)               |
|                                                                                                                                                                                             | -         | OFF       | -         | -         | Korea/Japan<br>Fire operation (JIS)         |
|                                                                                                                                                                                             | -         | <b>ON</b> | -         | -         | South-east Asia<br>Fire operation (BS)      |
|                                                                                                                                                                                             | -         | -         | OFF       | -         | ATT operation<br>type1 (ATT1) – (Note1)     |
|                                                                                                                                                                                             | -         | -         | <b>ON</b> | -         | ATT operation<br>type2 (ATT2) – (Note2)     |
|                                                                                                                                                                                             | -         | -         | -         | OFF       | Without NP,EP,MGR,RTN<br>Function           |
|                                                                                                                                                                                             | -         | -         | -         | <b>ON</b> | With NP,EP,MGR,RTN<br>Function (6~105m/min) |

Note1. ATT Operation type1 (ATT1) – When using ATT,AUP,ADOWN switch.

Note2.ATT Operation type2 (ATT2) – When using ATT,AUP,ADOWN,ASTART,  
APASS,ACUT switch.

|                                                                                   |                            |          |                  |
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## 6-2. Inverter auto-tuning and manual operation

3 phase line by motor, when the signal line is connected from Encoder to controller, the step of running the MMR controller with motor as following.

The driving is proceeded as below 10 steps.

- 1<sup>st</sup> : Inverter type Input  
↓
- 2<sup>nd</sup> : Input of Motor and Encoder information  
↓
- 3<sup>rd</sup> : Power re-input  
↓
- 4<sup>th</sup> : Lock  
↓
- 5<sup>th</sup> : Power re-input  
↓
- 6<sup>th</sup> : check the motor noise  
↓
- 7<sup>th</sup> : Input the speed gain  
↓
- 8<sup>th</sup> : Manual operation

### 1 step: Input the inverter type.

Input the value as following the CPM 10 of [SETUP] – [CPM] by OPP-2000.

In the case of 400v, 22kw inverter input '10' in this parameter.

| Parameter | Function      |          |         | Initial value | Range  | Unit | Effectiveness |
|-----------|---------------|----------|---------|---------------|--------|------|---------------|
| CPM10     | Inverter type |          |         | 7             | 1 ~ 10 | -    | 1             |
|           | Set value     | Capacity | Voltage |               |        |      |               |
|           | 1             | 5.5Kw    | 200V    |               |        |      |               |
|           | 2             | 7.5Kw    |         |               |        |      |               |
|           | 3             | 11Kw     |         |               |        |      |               |
|           | 4             | 15Kw     |         |               |        |      |               |
|           | 5             | 22Kw     |         |               |        |      |               |
|           | 6             | 5.5Kw    | 400V    |               |        |      |               |
|           | 7             | 7.5Kw    |         |               |        |      |               |
|           | 8             | 11Kw     |         |               |        |      |               |
|           | 9             | 15Kw     |         |               |        |      |               |
|           | 10            | 22Kw     |         |               |        |      |               |

|                                                                                   |                            |          |                  |
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## 2 step: Input the Motor and Encoder Value

Input the Motor and Encoder information.

In the case of motor data, input the parameter value on [SETUP] – [Set Motor] by OPP-2000 as reference the Motor and Encoder name plate.

| Parameter     | Function                     | Initial value | Range | Unit   |
|---------------|------------------------------|---------------|-------|--------|
| Motor RPM     | Input Motor rpm.             | 1450          | rpm   | 1      |
| Encoder Pulse | Input the Encoder Pulse no.  | 1024          | -     | 1      |
| Running Speed | Input the running speed/min. | 60            | M/sec | Direct |

1: After re-set or power re-input

## 3 step: After power OFF, Re-input.

By input the power re-input, it can make the input parameter on 2 step, validity.

## 4 step: Lock.

Refer to Appendix.

## 5 step: After power OFF and re-input.

By input the power re-input, automatically set parameter of CPM17, CPM18, CPM19 and CPM20 make validity.

## 6 step: Check the motor noise

After order 0(zero) speed on the inverter, check the motor noise.

To operate 0 speed, there is a way to operate after set up the manual speed 0.

In that case, input the rpm, 0 on the SPD3 of [SETUP] – [Speed RPM] by using the OPP-2000.

If there isn't noise on the motor after order 0 speed on the inverter with brake open, the value of

auto-tuning is O.K, but if there is big noise, it means the auto-tuning is failed. In that case, you must adjust the parameter of no. CPM17, CPM18, CPM19, CPM20 by manual tuning instead of Auto-Tuning. If there is no noise, go to the next 7<sup>th</sup> step.



### 7 step: Speed gain input

Reforming to the lift operation status, adjust properly parameter CPM 41~CPM 45 not so as speed over shute or under shute.

### 8 step: Manual Inspection

Input the manual rpm on SPD3 of [SETUP] – [Speed RPM] by using OPP-2000.

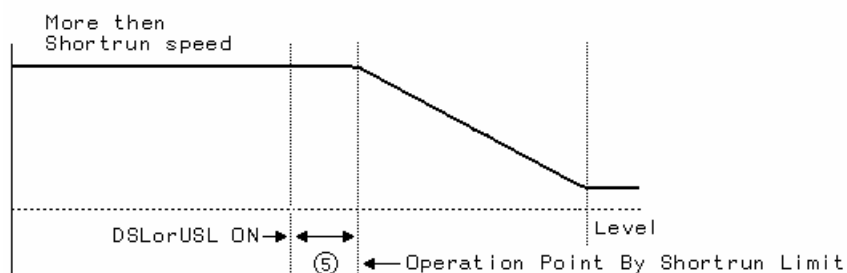
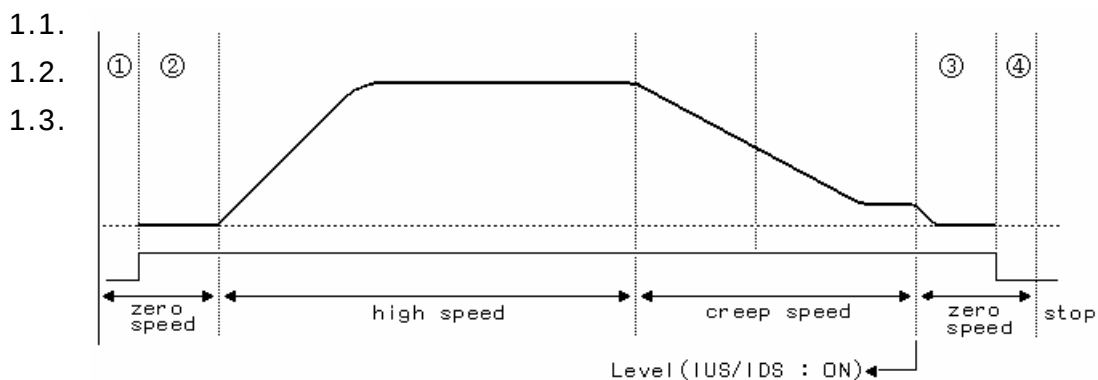
Furthermore, set up the manual deceleration time on CPM15 of [SETUP] – [CPM].

By using the manual operation(average 250 rpm), adjust the motor brake open/close time. Set up the motor brake/close time on [SETUP] – [IPM] by using OPP-2000.

| Parameter | Display | Function                                           |
|-----------|---------|----------------------------------------------------|
| IPM0      | -       | Motor Magnetization time (①)                       |
| IPM1      | -       | Brake opening time (②)                             |
| IPM2      | -       | Deceleration time to zero speed at Creep speed (③) |
| IPM3      | -       | Brake closing time (④)                             |



Do not get on top of the car before you confirm that there is no problem in operation in the machine-room.



|                                                                                   |                            |          |                  |
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### 6-3. Calibration

#### 6-3-1. Before calibration

- ① It should be Installed the Plate, IUS, IDS, DZONE(if you use) at each floor.
- ② DOOR open/close should be operated well.
- ③ Limit S/W should be installed (UL,USL,DL,DSL etc.).
- ④ Set up the limit contact type by dip s/w on the SMCB-3000Ci Board.
- ⑤ 4 of SW1 is OFF, Limit contact Type is “B” contact, and ON is “A” contact type.
- ⑥ Check whether the Limit S/W is installed well by Manual operation.
- ⑦ Check whether the [SETUP] – [Speed RPM]의 Zero Speed, Creep Speed, Rundown Speed, Calibration Speed are installed well by OPP-2000.
- ⑧ To operate other high speed, please do the calibration after checking all of things.



**NOTICE: UL or DL must not be input at the level of highest or lowest Floor.**

#### 6-3-2. Calibration

- ① Enter to [MODE] – [Calibration] by using OPP-2000. At this condition, when you press the [ENT] key, it will ask you whether you will do the calibration with [Calibration?] message. If you press the [ENT], it starts the calibration.
- ② When the calibration starts, it makes the car go to the DL.
- ③ When the car goes up and through DL to the lowest level, it is running with Leveling speed.
- ④ If it passes the lowest floor level, it is running to USL with calibration speed.
- ⑤ If the USL is input, the running decelerates and UP driving to UL with Leveling speed.
- ⑥ If the UL is input, it stops and store all of calibration data into the memory.
- ⑦ After that it changes to Auto-operation and moves to the top floor.
- ⑧ During calibration, if there the error is detected, it stops and display the Error code.
- ⑨ After solve the reason of Error and do the calibration again.

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### 6-3-3. After calibration, OPP setting

- ① Properly set up the below items by using OPP-2000.
- ② Set up the floor display and Indicator Lamp on [DISP].
- ③ Adjust the point of arriving gong, floor display changing, and deceleration distance on PPM of [SETUP]. You can adjust with running by OPP AUTO mode.
- ④ When it is OPP AUTO operation mode, you select OPP Auto Door of [OPT], NO and make door not open.
- ⑤ Properly set up the other items of [SETUP].  
For detail explanation for each item, refer to the OPP-2000 manual.
- ⑥ Properly set up the items of [OPT].  
For detail explanation for each item, refer to the OPP-2000 manual.
- ⑦ In the case of deceleration S/W is On at the top or lowest floor, check whether it properly decelerate or not. In that case or emergency stop, deceleration time is separately set up to CPM16.

### 6-4. Auto-Operation

- ① For Auto Operation there are OPP AUTO, AUTO REPEAT which can be operated by Car/Hall call operation and OPP-2000.
- ② Adjust the CPM0 ~ CPM8 of Acceleration / Deceleration time parameter as depends on Ride Comfort with Auto-operation.
- ③ In the case of parameter CPM0 ~ CPM8 of Acceleration / Deceleration time is changed, re-set up the PPM0 ~ PPM7.

---

<sup>1</sup> Depends on the running feel on the auto-operation, it can need to be adjusted the speed gain of 7<sup>th</sup> step.

## 2. OPP-2000 (Operator)

### 1. OPP-2000

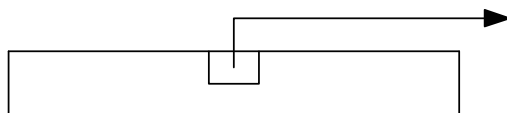
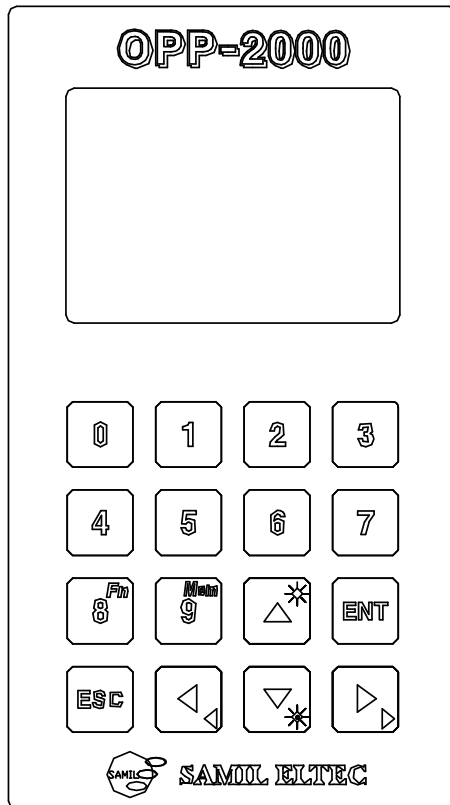
#### 1-1. Character

This product, OPP-2000, is simple and portable control unit which is able to monitoring of elevator running status, changing and fixing of operation mode and the value of elevator controlling data, and simply substituted by computer CRT.

Furthermore, it can control the NEW DSS, and MMR.

|                                                                                   |                            |          |                  |
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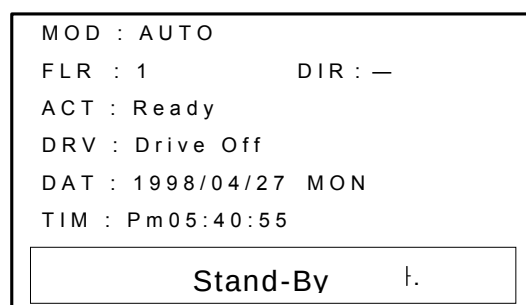
## 1-2. OPP-2000 Lay-Out



Communication Jack for Main Board  
(Or GCB Board)

## 2. OPP-2000 Menu

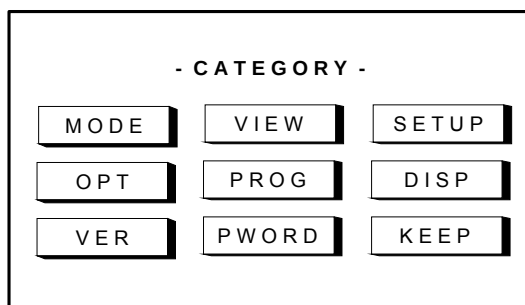
### 2-1. Main Menu



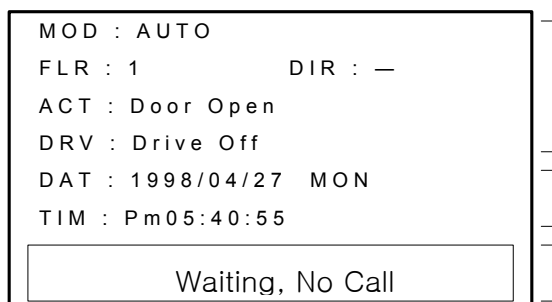
Push [ENT] Key for over 2sec, then Main menu is changed (converted to Category Menu.)

|                                                                                   |                            |          |                  |
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## 2-2. Category Menu



\*On the menu of OPP-2000 [ENT] Key means fixing of the data, and [ESC] key means cancel of the menu and convert back to original menu



|                                                                                   |                            |          |                  |
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### 3. OPP-2000 Menu

#### 3-1. Main Menu

Fixed menu Window

Changeable Menu Window

Message Window

1) MOD: Operation Mode of Elevator.

- STARTUP : Preparation of System Start.
- INSPECTION : Manual Operation Mode in out side.
- OPP INSPECTION : Manual Operation Mode by OPP-2000.
- CALIBRATION : Floor Height Check mode.
- PARKING : PARKING Mode.
- AUTO : Automatic operation Mode.
- OPP AUTO : Automatic Operation mode by OPP-2000.
- AUTO REPEAT : Automatic repeat Operation Mode.
- INDEPENDENT : Independent Operation Mode
- ATTENDANT : Attendant Operation Mode.
- FIREMAN CONTROL : Fireman Operation Mode.
- EACH FLOOR STOP : Each-Floor-Stop Operation Mode.
- FIRE CONTROL : Firing Control Operation Mode.
- SEISMIC VERY LOW : Seismic Control Operation Mode.
- SEISMIC LOW : Seismic Control Operation Mode.
- SEISMIC HIGH : Seismic Control Operation Mode.
- UPS CONTROL : UPS Control Operation Mode.
- GENERATOR CONTROL: Generator Operation Mode.
- MOVING HOUSE : Moving Operation Mode.
- OIL TEMP LOW : Checking sensor of Low Temp oil  
Operation mode.
- OIL TEMP HIGH : Checking sensor of Low Temp oil  
Operation mode.
- LOCK : Lock Mode
- RESET : System reset.

2) FLR: present Position of Car.

|                                                                                   |                            |          |                  |
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3) DIR : Operation Direction of Car, and the stop is ' \_ '

4) ACT: Operating status and Error message of Elevator.

- System Check : On checking system
- Not Calibration : On non-calibration

It should be calibration at the Auto driving

- Run Up : On calibrating of Floor Height
- Run Down : On Moving of Car to the lowest floor
- Run Down DL : On Moving of Car to DL.
- Leveling : On Moving of Car to the nearest floor.
- Auto Stop : On stopping with doors opening.
- Ready : On stopping with doors closing.
- No registered floor.
- Acceleration : On acceleration operation.
- High : On high speed operation.
- Deceleration : On deceleration operation.
- Inspection Up : On inspection operation with up direction.
- Inspection Down : On inspection operation with down

direction.

- Inspection Stop : On stopping in inspection operation.
- Door Open : On opening doors
- Door Close : On closing doors
- Door Stop SW On : Door stop switch is on.
- Emergency Stop : E-stop switch is on.
- Control Finished : On completion of parking, seismic, UPS and  
Emergency power operation.
- Group E-Stop : Emergency stop by Group controlling.
- Tuning Standby : Auto Tuning standby.
- Tuning Phase1 : On Auto Tuning 1<sup>st</sup> step.
- Tuning Phase2 : On Auto Tuning 2<sup>nd</sup> step.
- Tuning Phase3 : On Auto Tuning 3<sup>rd</sup> step.
- Tuning Waiting : Auto Tuning waiting
- Tuning Finish : Auto Tuning finish.
- Tuning Ready : Auto Tuning Ready.
- Tuning Failed : Auto Tuning Failed.

5) DRV: Indication of instruction for Elevator speed.

|                                                                                   |                            |          |                  |
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### 3-1-1. Changeable Menu Window

- 1) Changeable Menu window is indicating IN and OUT-PUT port and present status of each switch.
- 2) By pushing [ESC] key on main menu for 2second the window can be entered to the changeable menu selection mode, and then selecting the proposed menu using by [▲],[▼]key, and then entering into changeable Menu window. And then the status proposed can be shown using by [▲], [▼]key and [ENT] key.
- 3) The status in Changeable menu Window.
  - DAT : Date.
  - TIM : Time.
  - SPD : Max. Speed
  - RPM : Max. Motor RPM
  - PLS : Nos. of pulse for Car position.
  - DSW: Setting Value of Dip SW on MCU-2001Ci board.  
(Refer to SMCB-3000Ci Manual for Dip SW)

(Note) ON status ☐ ON, Off status ☐ OFF will be displayed.

- DOR
  - DOL: ON/OFF condition of Door Open Limit.
  - DCL: ON/OFF condition of Door Close Limit.
  - HCD: ON/OFF condition of Hall Door and Car Door Switch.
  - DOR: Indicating "ON" when doors are closed.
- IUD
  - IUS: ON/OFF condition of Inductor Up Switch.
  - IDS: ON/OFF condition of Inductor Down Switch
  - DOZ: Lighting "ON" when Car is on Door Zone.  
(ON/OFF condition of Door Zone Switch in case of using Door Zone Switch)
  - LEV: Indicating "ON" when Car is in Level.



|                                                                                   |                            |          |                  |
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- LT0
  - UL: Condition of Up Limit SW.
  - USL: Condition of Up slow Limit for Long Run operation.
  - DSL: Condition of Down Slow Limit for Long Run operation.
  - DL: Condition of Down Limit.
  
- LT1
  - HUS: Condition of Up Slow Limit for 3 Run operation
  - VUS: Condition of Up Slow Limit for 4 Run operation
  - VDS: Condition of Up Slow Limit for 4 Run operation
  - HDS: Condition of Up Slow Limit for 3 Run operation
  
- LT2 :
  - MUS: Condition of Up Slow Limit for 5 Run Operation.
  - SUS: Condition of Up Slow Limit for Short Run Operation.
  - SDS: Condition of Down Slow Limit for Short Run Operation
  - MDS: Condition of Down Slow Limit for 5 Run Operation.
  
- RSV :
  - PAK: Indicating "ON" for reserved parking operation.
  - SCH: Indicating "ON" for reserved schedule operation.
  - EFS: Indicating "ON" for reserved each floor stop operation.
  - NSF: Not Specified.
  
- OUT :
  - OPN: Indicating "ON" for Open Signal Output.
  - CLS: Indicating "ON" for Close Signal Output.
  - ACT: Indicating "ON" under operation.
  - BRK: Indicating "ON" for Brake Open
  
- DR2 :
  - EOP: Indicating "ON" for Open possible condition.
  - ECL: Indicating "ON" for Close possible condition.
  - F2D: Indicating "ON" for Output of Door Common signal during Door Common.
  - ALD: Indicating "ON" for output of Door Common signal during Leveling

|                                                                                   |                            |          |                  |
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- DR3 :
  - OP : ON/OFF condition for Open Switch.
  - HOP: ON/OFF condition for Holding Switch.
  - AOP: ON/OFF condition for disabled Open Switch.
  - SFT: ON/OFF condition for Safety Shoe switch.

- DR4 :
  - PTO: ON/OFF condition for photo switch
  - OVL: ON/OFF condition for Over Load switch.
  - DST: ON/OFF condition for Door Stop switch.
  - ETC: Not Specified.

\* "ON" for Data Indicating "I", "OFF" for "0"

|   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|
| ⑧ | ⑦ | ⑥ | ⑤ | ④ | ③ | ② | ① |
|---|---|---|---|---|---|---|---|

- IP0 : SMCB-3000Ci Board Input
  - ①: Indicating of input condition for INS Auto(1) / Inspection (0) switch.
  - ②: Indicating of input condition for Inspection Up switch.
  - ③: Indicating of input condition for Inspection Down switch.
  - ④: Indicating of input condition for inductor up switch.
  - ⑤: Indicating of input condition for Inductor Down switch.
  - ⑥: Indicating of input condition for Door Open Limit switch.
  - ⑦: Indicating of input condition for Door Close Limit switch.
  - ⑧: Not Specified
- IP1 : SMCB-3000Ci Board Input
  - ①: Indicating of Input condition for Hall and Car Door switch.
  - ②: Indicating of Input condition for Door Zone switch.
  - ③: Indicating of Input condition for MK1K2(Output Magnet Monitoring).
  - ④: Not Specified
  - ⑤: Indicating of Input condition for Up Limit switch.

|                                                                                   |                            |          |                  |
|-----------------------------------------------------------------------------------|----------------------------|----------|------------------|
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- ⑥: Indicating of Input condition for Up Slow Limit switch. (2 Run Up)
- ⑦: Indicating of Input condition for High Up Slow Limit switch.  
(3 Run Up)
- ⑧: Indicating of Input condition for Very High Up Slow Limit Switch.  
(4 Run Up)

- IP2 : SMCB-3000Ci Board Input

- ①: Indicating of Input condition for Most High Up Slow Limit switch.  
(5 Run Up)
- ②: Indicating of Input condition for Down Limit switch
- ③: Indicating of Input condition for Down Slow Limit switch.  
(2 Run Down)
- ④: Indicating of Input condition for High Down Slow Limit switch.  
(3 Run Down)
- ⑤: Indicating of Input condition for Very High Down Slow Limit switch. (4 Run Down)
- ⑥: Indicating of Input condition for Most High Down Slow Limit switch. (5 Run Down)
- ⑦: Indicating of Input condition for Parking switch.
- ⑧: Indicating of Input condition for Fireman operation Recall Switch.

- IP3 : SMCB-3000Ci Board Input

- ①: Emergency Stop
- ②: Inverter OC
- ③: UPS Power Supply Operation.
- ④: Brake Monitor Input.
- ⑤: Seismic (P Waive Input), Pre-Vibration.
- ⑥: Seismic (S Waive Input), Main-Vibration
- ⑦: Not Specified
- ⑧: Firing/ Fireman Operation switch

- OP0 : SMCB-3000Ci Board Output

- ①: Forward Control Signal (Up Direction)
- ②: Reverse Control Signal (Down Direction)
- ③: X1 Multi-Steps Speed Instruction
- ④: X2 Multi-Steps Speed Instruction
- ⑤: X3 Multi-Steps Speed Instruction

|                                                                                   |                            |          |                  |
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⑥: X4 Multi-Steps Speed Instruction (EXT1 Output)

⑦: No Use

⑧: No Use

- OP1 : SMCB-3000Ci Board Output

①: Open 1 Signal Output

②: Close 1 Signal Output

③: Active 1Signal Output

④: Close 2 signal output

⑤: Close 3 signal output

⑥: Close 4 signal output

⑦: No Usage for Output (No Port)

⑧: No Usage for Output (No Port)

- OP2 : SMCB-3000Ci Board output

①: ICUT signal output

②: Forced "ON" signal output for Doors.

③: BR1 signal output

④: BRCUT signal output

⑤: Removal Signal Output for Seismic Detector

⑥: DSOL signal output(Door Solenoid control output)

⑦: MAGNET signal output (Magnet control output)

⑧: DCOMR signal output

(Output the forcing common of door during using the Retreating CAM)

- BI0 : OPB-101 Board Input

①: Non-Identification

②: AIRCON signal Input (Air-condition sensor input)

③: PHOTO signal Input (Photo switch Input)

④: SAFETY signal Input (Safety Shoe switch Input)

⑤: OVERLOAD signal Input (Over Load switch Input)

⑥: FULL signal Input (Load Full(80%) Switch Input)

⑦: 45% Signal Input (Load 45% Switch Input)

⑧: 30% Signal Input (Load 30% Switch Input)

- BI1 : OPB-101 Board Input

①: ABZC signal Input (Attendant Buzzer Off switch)

|                                                                                   |                            |          |                  |
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- ②: APASS signal Input (Attendant Pass switch)
- ③: ASTART signal Input (Attendant Start switch)
- ④: ADONW signal Input (Attendant Down driving switch)
- ⑤: AUP signal Input (Attendant Up driving switch)
- ⑥: ATT signal input (Attendant driving select switch)
- ⑦: FIRE 2 signal Input (2<sup>nd</sup> fireman switch)
- ⑧: FIRE 1 signal switch (1<sup>st</sup> fireman switch)

- BI2 : OPB-101 Board input

- ①: DRST signal Input (Door forcing Stop switch)
- ②: BBSW signal Input (each-floor driving select switch)
- ③: MH Signal Input (House-moving select switch)
- ④: IND signal Input (Exclusive use select switch)
- ⑤: OPEN 2 signal Input (Handicap OPEN switch)
- ⑥: OPEN 1 signal Input (Open Switch)
- ⑦: CLOSE signal Input (Close switch)
- ⑧: OPEN signal Input (Open switch)

- BO0 : OPB-101 Board output

- ①: Not Specified
- ②: Not Specified
- ③: Not Specified
- ④: Not Specified
- ⑤: Not Specified
- ⑥: BF signal output (when the car stops at the lowest floor, 1)
- ⑦: RE signal output (When the FIREMAN RECALL) input switch is ON, 1)
- ⑧: FIR1 signal output (when the 1<sup>st</sup> fireman switch is ON, 1)

- BO1 : OPB-101 Board output

- ①: Buzzer Signal Output
- ②: Up Chime
- ③: Down Chime
- ④: Fan (Connected working with CP FAN)
- ⑤: Light
- ⑥: L41 signal output (Open button Lamp)
- ⑦: L42 signal output (Close button Lamp)
- ⑧: L43 signal output (handicap or open button Lamp)

|                                                                                   |                            |          |                  |
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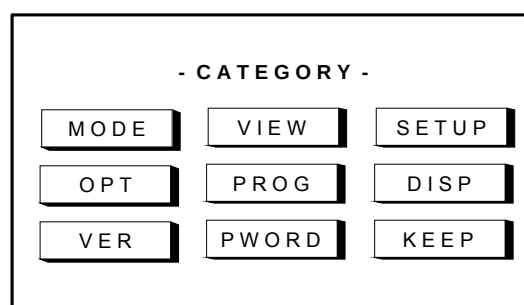
- DOB: An inverter digital output state indication.

- ①: Regeneration OFF, Magnet OFF
- ②: Regeneration OFF, Magnet ON
- ③: Regeneration ON, Magnet OFF
- ④: Regeneration ON, Magnet ON
- ⑤: Not Specified
- ⑥: Not Specified
- ⑦: Not Specified
- ⑧: Not Specified

- DIB : An inverter digital input state indication

- ①: A electrical part overheating detection sensor  
(0: OFF, 1: ON)
- ②: Regeneration OFF, Magnet ON
- ③: Regeneration ON, Magnet OFF
- ④: Regeneration ON, Magnet ON
- ⑤: Not Specified
- ⑥: Not Specified
- ⑦: Not Specified
- ⑧: Not Specified

### 3-2. Category Menu



|                                                                                   |                            |          |                  |
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### 3-2-1. Mode Menu

#### 1) AUTO

- Automatic operation Mode by Car or Hall Call.
- When the Mode is selected, it goes to the beginning menu.

#### 2) AUTO Repeat

- Operating between selected Floors repeatedly.
- According to 「OPT」 OPP Auto Door at "OPT", Door Open or Door Close.  
When you select the [Yes], the Door is open during driving, and [No] select is not open.

If you press [ENT] Key, can display level selection Menu and then select repeat operating floor.

< Auto Repeat menu >

Auto Repeat :

FLR : 2                      DIR : —

B2

B1

1

2

3

4

5

6

Use [UP] / [DOWN] Key

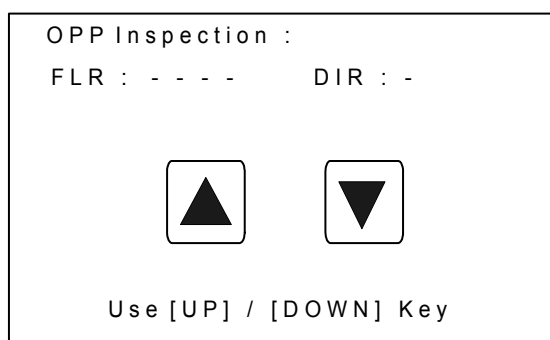
#### 3) OPP AUTO

- Operation Mode is OPP auto operation mode and ignore Car / Hall Call
- According to OPP Auto Door installation in 「OPT」, can choose whether door open or not. When you select the [Yes], the Door is open during driving, and [No] select is not open.
- If you press [ENT] Key, can display level selection Menu and then select repeat operating floor.

|                                                                                   |                            |          |                  |
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#### 4) OPP Inspection

- Selecting manual operation for operation Mode.  
(But, have to be Auto Operation Mode, invalidity in case of Manual Operation Manual)
- If press [ENT] Key on this Mode, OPP Inspection Menu will be displayed.
- Operating during pressing 「↑」, 「↓」 key Only.



< OPP Inspection menu >

#### 5) CALIBRATION

- Operating to measure the Height of floor.
- For detailed explanation, Please see the appendix 4-1.

#### 6) LOCK

- Do the Lock with Brake open.
- All of Calls are not registered and to be the inspection mode.
- For detailed explanation, Please see the appendix (X-X)

#### 7) RESET

- Reset the Inverter.
- The mode doesn't change even you operate the RESET.

### 3-2-2. View Menu

#### 1) Car / Hall Call

- Display Car / Hall Call registered status
- ◆ is the registered status of Hall Call UP/DOWN • is the registered status of Car Call.



|                                                                                   |                            |          |                  |
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## 2) Car / Hall Call AB

- handicapped Car or Hall Call registered
- ◆ is the registered status of Hall Call UP/DOWN • is the registered status of Car Call

## 3) Defect Button

- Indicating Hall call button which is pressed over pre-setting time at GPM 6.
- Not detect the Car Call button.

## 4) Defect Button AB

- Indicating Disabled button which is pressed over pre-setting time at GPM6.
- Not detect the Car Call button.

## 5) Connection IND

- Indicating floor which is connected with Indicator.

## 6) Service Stop SW

- Indicating condition for stoppage of the service ON/OFF.
- No service the set floor.

## 7) Running Time

- Indicating total operating time.

## 8) Distance

- Floor Height – Indicating distance between floors.
- Plate Length – Indicating length of all kinds of Plate.
- IUS-IDS – Indicating distance between IUS and IDS.
- Limit SW – Indicating position of each Limit.

### 9) Running Counter

- Display total operation times.

### 10) SPM (Only displayed at the MMR)

- Inverter Condition.

| Parameter | Display        | Function                                                                                                                                                                                                                                                                                                       |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------|-----|-----|-----|---|-----|----|---|----|-----|---|----|----|
| SPM0      | Target RPM     | Show the goal speed with rpm.                                                                                                                                                                                                                                                                                  |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM1      | Command RPM    | Show the Acceleration/deceleration with rpm                                                                                                                                                                                                                                                                    |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM2      | Current RPM    | Show the current speed of motor with rpm                                                                                                                                                                                                                                                                       |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM3      | Current Torque | The output current show with %.                                                                                                                                                                                                                                                                                |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM4      | Peak Torque    | The maximum output current by motor is showed by %                                                                                                                                                                                                                                                             |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM5      | Current        | Show the output current with present motor to RMS, the unit is 100mA.                                                                                                                                                                                                                                          |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM6      | Voltage        | Show the output current value by present motor to RMS, the unit is V                                                                                                                                                                                                                                           |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM7      | DC BUS Voltage | Show the Voltage of DC bus of inside inverter electric part, the unit is V.                                                                                                                                                                                                                                    |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM8      | DO Status      | Show the condition of digital output of Inverter, the meaning of value is below that.                                                                                                                                                                                                                          |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
|           |                | <table><tr><td>Display</td><td>Resurrection</td><td>Protection resistance from input current of part Bypass Magnetic</td></tr><tr><td>0</td><td>off</td><td>Off</td></tr><tr><td>1</td><td>off</td><td>on</td></tr><tr><td>2</td><td>on</td><td>off</td></tr><tr><td>3</td><td>on</td><td>on</td></tr></table> | Display                               | Resurrection                                                     | Protection resistance from input current of part Bypass Magnetic | 0   | off | Off | 1 | off | on | 2 | on | off | 3 | on | on |
|           |                | Display                                                                                                                                                                                                                                                                                                        | Resurrection                          | Protection resistance from input current of part Bypass Magnetic |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
|           |                | 0                                                                                                                                                                                                                                                                                                              | off                                   | Off                                                              |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
|           |                | 1                                                                                                                                                                                                                                                                                                              | off                                   | on                                                               |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
|           |                | 2                                                                                                                                                                                                                                                                                                              | on                                    | off                                                              |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| 3         | on             | on                                                                                                                                                                                                                                                                                                             |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| SPM9      | DI Status      | Show the condition of digital input of Inverter, the meaning of value is below that.                                                                                                                                                                                                                           |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
|           |                | <table><tr><td>Display</td><td>Sensor for over heat of electric part</td></tr><tr><td>0</td><td>Off</td></tr><tr><td>1</td><td>On</td></tr></table>                                                                                                                                                            | Display                               | Sensor for over heat of electric part                            | 0                                                                | Off | 1   | On  |   |     |    |   |    |     |   |    |    |
|           |                | Display                                                                                                                                                                                                                                                                                                        | Sensor for over heat of electric part |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| 0         | Off            |                                                                                                                                                                                                                                                                                                                |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |
| 1         | On             |                                                                                                                                                                                                                                                                                                                |                                       |                                                                  |                                                                  |     |     |     |   |     |    |   |    |     |   |    |    |



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**FOD**

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## Installation Manual

SPM10

Inv.  
Fault

Show the alarm message of Inverter with number. Depends on the value, the meaning is below that

| No. | Alarm Message      | No. | Alarm Message   |
|-----|--------------------|-----|-----------------|
| 0   | No Message         | 9   | Motor Parameter |
| 1   | IPM Fault          | 10  | Low Power       |
| 2   | Over Voltage       | 11  | Encoder Input   |
| 3   | Under Voltage      | 12  | Over Heat       |
| 4   | Motor Overload     | 13  | Watchdog INV    |
| 5   | Over Speed         | 14  | Not Specified   |
| 6   | Excess Speed Error | 15  | Not Specified   |
| 7   | DSP Fault1         | 16  | Not Specified   |
| 8   | DSP Fault2         | 17  | Not Specified   |

\*. Refer to 4-2-3 Inverter Encoder Code regarding the explanation of alarm Code.

| Parameter | Display        | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|-----------|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------|---------|---|------|--------------------------------------------------------|---|------|-------------------------------------------------------|---|---|------------------------------------------------|---|-------------------------------------------|---|------------------------------------|---|------------------------------------|---|------------------------------------|---|------------------------------------|---|--------------------------------------------|---|-------------------------------------------------------------|---|------------------------------------------------|----|----------------------------------------------------------------------------------------------|----|----------------------------------------|
| SPM11     | Inv. Sub Fault | Explains auxiliary condition situation of inverter alarm (SPM10).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           |                | <table border="1"> <tr> <th>SPM10</th><th>SPM11</th><th>Meaning</th></tr> <tr> <td>2</td><td>xxxx</td><td>When over voltage, Displaying DC bus voltage(Unit : V)</td></tr> <tr> <td>3</td><td>xxxx</td><td>When low voltage, Displaying DC bus voltage(Unit : V)</td></tr> <tr> <td rowspan="12">9</td><td>1</td><td>Constant Error converting ADC value to current</td></tr> <tr> <td>2</td><td>Constant Error converting Pulse to speed.</td></tr> <tr> <td>3</td><td>Constant Error for current, P Gain</td></tr> <tr> <td>4</td><td>Constant Error for current, I Gain</td></tr> <tr> <td>5</td><td>Constant Error for current, I Gain</td></tr> <tr> <td>6</td><td>Constant Error for No Load Current</td></tr> <tr> <td>7</td><td>Constant Error for instructing Max Torque.</td></tr> <tr> <td>8</td><td>Over Flow of Current measurement or Excessive Current Range</td></tr> <tr> <td>9</td><td>Excessive Range of over Inductance Measurement</td></tr> <tr> <td>10</td><td>When synchronous speed is less than nominal speed, or Excessive Time constant Range of Rotor</td></tr> <tr> <td>11</td><td>Excessive Time Constant Range of Rotor</td></tr> <tr> <td>12</td><td>Time out when measurement of Excessive Inductance</td></tr> </table> | SPM10 | SPM11 | Meaning | 2 | xxxx | When over voltage, Displaying DC bus voltage(Unit : V) | 3 | xxxx | When low voltage, Displaying DC bus voltage(Unit : V) | 9 | 1 | Constant Error converting ADC value to current | 2 | Constant Error converting Pulse to speed. | 3 | Constant Error for current, P Gain | 4 | Constant Error for current, I Gain | 5 | Constant Error for current, I Gain | 6 | Constant Error for No Load Current | 7 | Constant Error for instructing Max Torque. | 8 | Over Flow of Current measurement or Excessive Current Range | 9 | Excessive Range of over Inductance Measurement | 10 | When synchronous speed is less than nominal speed, or Excessive Time constant Range of Rotor | 11 | Excessive Time Constant Range of Rotor |
| SPM10     | SPM11          | Meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
| 2         | xxxx           | When over voltage, Displaying DC bus voltage(Unit : V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
| 3         | xxxx           | When low voltage, Displaying DC bus voltage(Unit : V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
| 9         | 1              | Constant Error converting ADC value to current                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 2              | Constant Error converting Pulse to speed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 3              | Constant Error for current, P Gain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 4              | Constant Error for current, I Gain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 5              | Constant Error for current, I Gain                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 6              | Constant Error for No Load Current                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 7              | Constant Error for instructing Max Torque.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 8              | Over Flow of Current measurement or Excessive Current Range                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 9              | Excessive Range of over Inductance Measurement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 10             | When synchronous speed is less than nominal speed, or Excessive Time constant Range of Rotor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 11             | Excessive Time Constant Range of Rotor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |
|           | 12             | Time out when measurement of Excessive Inductance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |       |         |   |      |                                                        |   |      |                                                       |   |   |                                                |   |                                           |   |                                    |   |                                    |   |                                    |   |                                    |   |                                            |   |                                                             |   |                                                |    |                                                                                              |    |                                        |

|       |                |                                                                               |                       |
|-------|----------------|-------------------------------------------------------------------------------|-----------------------|
|       |                | 14                                                                            | A slip constant error |
|       |                |                                                                               |                       |
| SPM12 | -              | Display Load Sensor Value/Toque Bias<br>An indication from a more than OPP2.1 |                       |
| SPM13 | Empty Load A   | PTM A sensor value taught with empty car,<br>0% button on OPB board           |                       |
| SPM14 | Empty Load B   | PTM B sensor value taught with empty car,<br>0% button on OPB board           |                       |
| SPM15 | Full Load A    | PTM A sensor value taught with full car,<br>100% button on OPB board          |                       |
| SPM16 | Full Load B    | PTM B sensor value taught with full car,<br>100% button on OPB board          |                       |
| SPM17 | Current Load A | Current PTM A sensor value                                                    |                       |
| SPM18 | Current Load B | Current PTM B sensor value                                                    |                       |

## 11) Warning Report

- Record works when a warning Error occurrence.
- Please refer to Error Code (4-1).

## 12) Fault Report

- Error Code becomes a record when trouble occurrence
  - Please refer to Error Code (4-1).

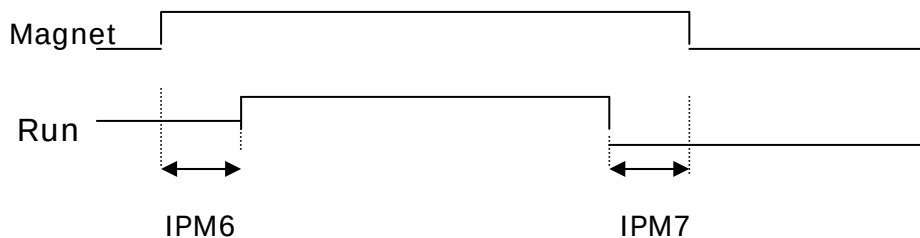
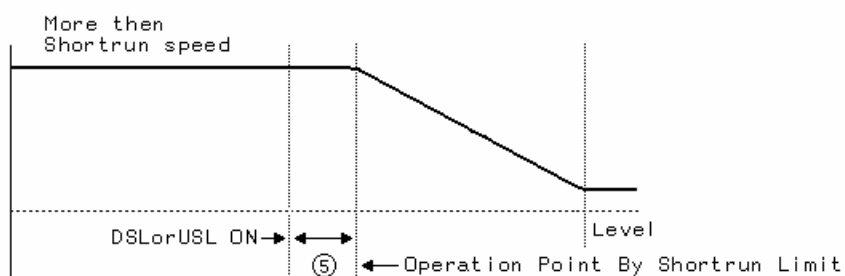
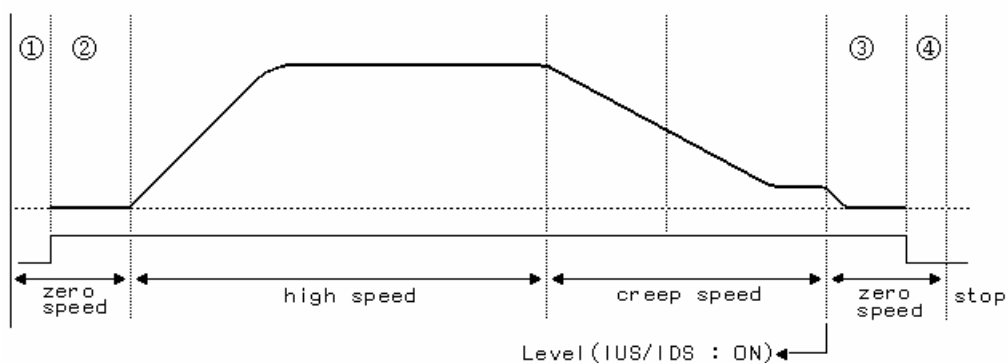
|                                                                                   |                            |          |                  |
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### 3-2-3. Setup Menu

- It will be changing mode after moving to the arrow Key and press the [ENT] KEY
- When you input the value by No. key and press the [ENT] KEY, it shows the message which asks the check of changing. When you select the Yes[ENT], it memorize, but No[ESC] doesn't memorize.

#### 1) IPM

| Parameter | Display            | Function                                                                                            |
|-----------|--------------------|-----------------------------------------------------------------------------------------------------|
| IPM0      |                    | Motor Magnetic Time (①)                                                                             |
| IPM1      |                    | Brake open time (②)                                                                                 |
| IPM2      |                    | Deceleration time from Creep speed to Zero speed (③)                                                |
| IPM3      |                    | Brake close time (④)                                                                                |
| IPM4      | SDSL<br>Limit Time | Decide the point of time of SDSL after input the DSL (⑤)                                            |
| IPM5      | SUSL<br>Limit Time | Decide the point of time of SDSL after input the USL (⑤)                                            |
| IPM6      |                    | Time for magnet ON before Inverter drive<br>* Only possible when the Magnet Control option is [Yes] |
| IPM7      |                    | Magnet off time after Inverter stop (note)efficiency when the Magnet Control option is [Yes]        |



### 2) DPM

· Setting the Parameter for door.

| Parameter | Display          | Function                                                                                                                                       |
|-----------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| DPM0      | DT Normal        | Door close time, 0 is Open waiting                                                                                                             |
| DPM1      | DT Hold Button   | Close time when the open button operates, 0 is Open waiting                                                                                    |
| DPM2      | DT Abnormal      | handicap Door close time, 0 is Open waiting                                                                                                    |
| DPM3      | DS Normal        | Door open/close signal output time (msec)                                                                                                      |
| DPM4      | DS After DCL in  | Close signal output time after DCL input(msec)                                                                                                 |
| DPM5      | DS After DOL in  | Open signal output time after DOL input (msec)                                                                                                 |
| DPM6      | Door Change Time | Delay time when Door Open / Close reverse (msec)                                                                                               |
| DPM7      | Buzzer Time      | Close forecast buzzer output time when the open button operates<br>Buzzer output time before Door close                                        |
| DPM8      | Open Detect Time | When door is open during driving, detect time with door open(m/sec). When the door is open over setting time, the car stop and rescue driving. |
| DPM9      | DT Special       | Door close time during earthquake, power off, fire                                                                                             |

### 3) PPM

| Parameter | Display            | Function                        |
|-----------|--------------------|---------------------------------|
| PPM0      | Display Change     | Floor or No. changing time      |
| PPM1      | Arrival Gong       | Arrival gong time               |
| PPM2      | Short run Point    | Short run deceleration distance |
| PPM3      | Long run Point     | Long run deceleration distance  |
| PPM4      | Three run Point    | 3 run deceleration distance     |
| PPM5      | Four run Point     | 4 run deceleration distance     |
| PPM6      | Five run Point     | 5 run deceleration distance     |
| PPM7      | Running Open Point | Running Open point              |



### 4) FPM

| Parameter | Display           | Function                                                                                                                                                                                                                                                                                                    |
|-----------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FPM0      | Company Code      | Company code input.<br>No company Code, set to 0.                                                                                                                                                                                                                                                           |
| FPM1      | Hydraulic Type    | 0 : EV-100 control output<br>1 : Saturn Alfa<br>(Inverter Type Power Unit) control                                                                                                                                                                                                                          |
| FPM2      | Leveling Type     | 0: If there is direction, no need Leveling.<br>Up leveling of hydraulic can be possible after getting rid of IDS and DZONE.<br>1: Leveling doesn't concern about direction<br>Lamp and can do the leveling after getting rid of IUS and IDS at the hydraulic                                                |
| FPM3      | Parking Type      | 0: Parking after service the hall call and car call.<br>1: Ignore the Hall Call, and parking after service the Car Call.                                                                                                                                                                                    |
| FPM4      | Door Time Type    | 0 : DPM0 – Normal Door Time<br>DPM1 – Open Button Door Time<br>1 : DPM0 – Car Call Door Time<br>DPM1 – Hall Call Door Time                                                                                                                                                                                  |
| FPM5      | Door Close Buzzer | 0 : There not being a buzzer output function in J23-8 of OPB-101<br>1 : Buzzer output for I see before closing door at OPB-101 J23-8                                                                                                                                                                        |
| FPM6      | Door Open Type    | When landing, conditional selection of door open<br>0: Open the door only when Car/Hall call is registered at landing Floor.<br>1: Open the door regardless of Car/Hall call at landing floor, when condition is acceptable to open the door.<br>*. This is not applicable for OPP AUTO & AUTO REPEAT Mode. |

|                                                                                   |                            |          |                  |
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|      |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FPM7 | Port<br>Function  | Output selection of SMCB-3000Ci CN13-13<br>0 : EQRST<br>(Reset output for seismic sensor)<br>1 : Down driving direction Lamp output<br>2 : displayed when full<br>3 : displayed when during a check<br>4 : displayed when automatic mode<br>5: Output for over passenger capacity.<br>6 : displayed when driver operation mode<br>7 : displayed when only driving mode<br>8: Output for Fire operation mode.<br>9 : Output for Parking operation mode<br>10 : Output for Seismic control<br>11 : Output for Power failure control<br>12 : Output for Fire control<br>13 : Output for Fire control<br>14 : Output for Fire control<br>15 : Suspense<br>16 : Output for Moving operation<br>17: Output for Normal operation Available.<br>18 : displayed when trouble<br>19 : Output for BS 2 <sup>nd</sup> Fire operation<br>20 : Output for Generator operation<br>21 : Output for Cease of control operation<br>22 : Output under operation<br>23 : Output under Stand still |
| FPM8 | Next<br>Direction | When landing, selection of next operational direction decision timing<br>0 : Next operational direction decision when the timing of Gong<br>1 : Next operational direction decision when door open at landing (Set 1, when the display PCB is DOT-Matrix)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

5) GPM

| Parameter | Display          | Function                                                                                                                                                                                       |
|-----------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GPM0      | Power On Delay   | Delay time of driving order after Power On                                                                                                                                                     |
| GPM1      | Over run Time    | Maximum driving time for 1 floor                                                                                                                                                               |
| GPM2      | Fan Off Time     | Automatically fan off time inside the car                                                                                                                                                      |
| GPM3      | Light Off Time   | Automatically light off time inside the car                                                                                                                                                    |
| GPM4      | Schedule Time    | Changing time to Schedule driving<br>Changing time to schedule driving if there is no Car or Hall Call<br>(Note.) It is available during schedule driving                                      |
| GPM5      | Seismic Reset    | Seismic Reset time.                                                                                                                                                                            |
| GPM6      | Button Hold Time | Searching time of Faulty Button.<br>When the button is jammed over setting time of GPM6, it ignores the registered function.<br>(Note.) It is available when the defect button option is [Yes] |
| GPM7      | Hall Lantern     | Hall Lantern off.<br>Setting the flicking time of direction lamp.<br>(Note.) It is available when the Hall Lantern option is [Yes]                                                             |
| GPM8      | ATT Buzzer Time  | Buzzer time when the door is open at the ATT mode and the hall call is registered at another floor. Cancel the buzzer function when you set 0<br>(Note.) It is available at the ATT mode.      |
| GPM9      | Brake1 Off Time  | Setting the off time of Brake1 signal.                                                                                                                                                         |
| GPM10     | Nudging Time     | Setting the Door Nudging operating time. Set up to 0, Nudging function is ignored.                                                                                                             |
| GPM11     | Low Temp Time    | At the hydraulic type, when low temperature sensor is working, circle running time                                                                                                             |
| GPM12     | Solenoid On      | Before Door Open, Solenoid On time.<br>(Note.) Door Solenoid option is [Yes]                                                                                                                   |
| GPM13     | Solenoid Off     | After Open, Set up the Solenoid Off point.<br>When you set up to 0, Solenoid is On during Door open.<br>(Note.) Door Solenoid option is [Yes]                                                  |
| GPM14     | CRT Watchdog     | Time to check between CRT and communication.                                                                                                                                                   |
| GPM15     | Reopen Door Time | Applicable separately for door time when landing & reopen<br>0 : Not intended.<br>ETC: When reopen, set by door time.                                                                          |

|                                                                                   |                            |          |                  |
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#### 6) CPM

- Setting the parameter for control of Inverter.
- When the value Inputs less than minimum input value, it automatically changes to minimum value, and input more than maximum value, it changes to maximum value.
- CPM Parameter can be useful after changing directly, or reset or power in put again

1: After reset or power input again

| Parameter | Functional Explanation                              |                                                                                                                     | Initial Value | Range     | Unit | Effectiveness |
|-----------|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------|-----------|------|---------------|
| CPM0      | S curve Acceleration Start Area, %                  | Applicable for all Acceleration Except Max Speed.<br>(In case of 1 floor operation Applicable for all Acceleration) | 50            | 0 ~ 50    | %    | Immediately   |
| CPM1      | S curve Acceleration Linear Area, %                 |                                                                                                                     | 200           | 100 ~ 500 | 10ms | Immediately   |
| CPM2      | S curve Acceleration Final Area, %                  |                                                                                                                     | 35            | 0 ~ 50    | %    | Immediately   |
| CPM3      | S curve 1 <sup>st</sup> Deceleration Start Area, %  | Applicable for Creep speed Deceleration at all high speed Except Max Speed.                                         | 20            | 0 ~ 50    | %    | Immediately   |
| CPM4      | S curve 1 <sup>st</sup> Deceleration Linear Area, % |                                                                                                                     | 200           | 100 ~ 500 | 10ms | Immediately   |
| CPM5      | S curve 1 <sup>st</sup> Deceleration Final Area, %  |                                                                                                                     | 40            | 0 ~ 50    | %    | Immediately   |
| CPM6      | S curve Creep Deceleration Start Area, %            | Applicable for Deceleration at Creep or                                                                             | 0             | 0 ~ 50    | %    | Immediately   |

|       |                                                       |                                                                                                        |      |              |       |             |
|-------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------|------|--------------|-------|-------------|
| CPM7  | S curve<br>Creep<br>Deceleration<br>Linear<br>Area, % | leveling<br>speed.                                                                                     | 1000 | 100 ~<br>500 | 10ms  | Immediately |
| CPM8  | S curve<br>Creep<br>Deceleration<br>Final<br>Area, %  |                                                                                                        | 15   | 0 ~<br>50    | %     | Immediately |
| CPM9  | OSE TIME                                              | When losing<br>speed and it<br>is less than<br>this time,<br>Automatically<br>neglect<br>losing Alarm. | 0    | 0 ~<br>1000  | 10ms  | Immediately |
| CPM10 | Inverter form                                         |                                                                                                        | 7    | 1 ~<br>11    | -     | 1           |
|       | Fixed<br>value                                        | Capacity                                                                                               |      |              |       |             |
|       | 1                                                     | 5.5Kw                                                                                                  |      |              |       |             |
|       | 2                                                     | 7.5Kw                                                                                                  |      |              |       |             |
|       | 3                                                     | 11Kw                                                                                                   |      |              |       |             |
|       | 4                                                     | 15Kw                                                                                                   |      |              |       |             |
|       | 5                                                     | 22Kw                                                                                                   |      |              |       |             |
|       | 6                                                     | 5.5Kw                                                                                                  |      |              |       |             |
|       | 7                                                     | 7.5Kw                                                                                                  |      |              |       |             |
|       | 8                                                     | 11Kw                                                                                                   |      |              |       |             |
|       | 9                                                     | 15Kw                                                                                                   |      |              |       |             |
|       | 10                                                    | 22Kw                                                                                                   |      |              |       |             |
| CPM11 | Encoder Direction Change                              |                                                                                                        | 0    | 0 ~ 1        | -     | 1           |
| CPM12 | Debug Item (Not specified)                            |                                                                                                        | 1    | 0 ~<br>100   | -     | Immediately |
| CPM13 | No Use                                                |                                                                                                        | -    | -            | -     | -           |
| CPM14 | No Use                                                |                                                                                                        | -    | -            | -     | -           |
| CPM15 | Manually<br>Linear Area.                              | Deceleration                                                                                           | 150  | 100 ~<br>500 | 10ms- | Immediately |

| Parameter | Functional Explanation                           |                                                                                                                                                          | Initial Value | Range    | Unit | Effectiveness |
|-----------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------|------|---------------|
| CPM38     | Empty Load Bias %                                | Torque Bias Value applicable in Empty load (Value is based in normal 100%)                                                                               | 50            | 30 ~ 60  | %    | Immediately   |
| CPM39     | Full Load Sensor Value                           | Input reading value, SMP12 Load value from Load Sensor at Full Load.                                                                                     | 1023          | 0 ~ 1023 | -    | Immediately   |
| CPM40     | Full Load Bias %                                 | Torque Bias Value applicable in Empty load (Value is based in normal 100%)                                                                               | 50            | 30 ~ 60  | %    | Immediately   |
| CPM41     | Gain Changeable Speed                            | Use Gain-2 when speed over than specified in CPM41. When speed less than that, decide the Gain according to the straight line between Gain 1 and Gain 2. | 0             | 0 ~ 100  | %    | Immediately   |
| CPM42     | Speed Loop P Gain-1                              |                                                                                                                                                          | 150           | 1 ~ 1200 | -    | Immediately   |
| CPM43     | Speed Loop I Gain-1                              |                                                                                                                                                          | 20            | 0 ~ 2100 | -    | Immediately   |
| CPM44     | Speed Loop P Gain-2                              |                                                                                                                                                          | 150           | 1~1200   | -    | Immediately   |
| CPM45     | Speed Loop I Gain-2                              |                                                                                                                                                          | 20            | 0 ~ 2100 | -    | Immediately   |
| CPM46     | Running Open RPM                                 | When Running open, Set RPM permissible for Door Open.                                                                                                    | 0             | 0~9999   | rpm  | Immediately   |
| CPM47     | It's engineering parameter.<br>9 = MMR for SIGMA |                                                                                                                                                          | 1             | 0~9999   | -    | 1             |
| CPM48     | lock data13                                      |                                                                                                                                                          | 2867<br>2     | 0~65535  | -    | 1             |
| CPM49     | lock data32                                      |                                                                                                                                                          | 2867<br>2     | 0~65535  | -    | 1             |
| CPM50     | lock data26                                      |                                                                                                                                                          | 2867<br>2     | 0~65535  | -    | 1             |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |         |   |   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---------|---|---|
| CPM51 | lock data64                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2867<br>2 | 0~65535 | - | 1 |
| CPM52 | lock data45                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 2867<br>2 | 0~65535 | - | 1 |
| CPM53 | lock data51<br>The above lock data is automatically updated when locking operation is successful. Thus they are never changed arbitrarily by user.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2867<br>2 | 0~65535 | - | 1 |
| CPM54 | IRIS1<br>10 : GTW2-81P0,(450~800kg,60m/min)<br>11 : GTW2-81P5,(450~800kg,90m/min)<br>12 : GTW2-81P7,(450~800kg,105m/min)<br>13 : GTW2-101P0,(900~1000kg,60m/min)<br>14 : GTW2-101P5,(900~1000kg,90m/min)<br>15 : GTW2-101P7,(900~1000kg,105m/min)<br>16 : GTN1-101P7,(450~1000kg,105m/min),<br><b>TR &gt; 60m</b><br>17 : GTN2-161P0,(1,150~1600kg,60m/min)<br>18 : GTN2-161P5,(1,150~1600kg,90m/min)<br>19 : GTN2-161P7,(1,150~1600kg,105m/min)<br>IRIS2<br>100 : YTW260-G2F,(800~1150kg,120m/min)<br>101 : YTW260-G2F,(800~1150kg,150m/min)<br>102 : YTW260-G2FQ,(1,350~1600kg,120m/min)<br>103 : YTW260-G2FQ,(1,350~1600kg,150m/min) | 0         | 0~255   | - | 1 |

|                                                                                   |                            |          |                  |
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7) No Door Floor

- No Door floor setting.
- Non stop even emergency driving.

8) Non Service Hall

- Setting the Hall Call service prohibited floor.
- No Hall Call at the setting floor.

9) Non Service Car

- Setting the Car Call service prohibited floor.
- No Car Call at the setting floor.
- Ignore during emergency driving.

10) Auto Service UP

- Even though there is no registration during up driving, setting the automatically stop floor.

11) Auto Service Down

- Even though there is no registration during up driving, setting the automatically stop floor

12) Set Odd Floor

- A selection does an odd number floor.
  - Only the floor that was designated to an odd number floor if you select an odd number floor driving mode holds service.
- Odd number floor driving mode input is J26-6 of OPB-101.

13) Set Even Floor

- A selection does an even number floor.
- Only the floor that was designated to an even number floor if you select an even number floor driving mode holds service.



|                                                                                   |                            |          |                  |
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Even number floor driving mode input is J26-7 of OPB-101

#### 14) A Door Floor ( It is useful at the version of two doors)

- Setting the Door floor.
- No door open at the no setting floor.

#### 15) B Door Floor (It is useful at the version of two doors)

- Setting the B Door floor
- No door open at the no setting floor

#### 16) Input Type MCU

- Input type setting of SMCB-3000 Board.
- “-“ is Normal Open as A contact “ B” is Normal Close as B contact.
- Black part is setting by Dip switch and no change here.
- Control bit is as below.

| Parameter | 7       | 6       | 5     | 4     | 3     | 2    | 1     | 0     |
|-----------|---------|---------|-------|-------|-------|------|-------|-------|
| SIP0      | Suspens | DCL     | DOL   | IDS   | IUS   | IDN  | IUP   | INS   |
| SIP1      | VHUSL   | HUSL    | USL   | UL    | DCLr  | DOLr | DZONE | DOOR  |
| SIP2      | RECALL  | PKS     | MHDSL | VHDSL | HDSL  | DSL  | DL    | MHUSL |
| SIP3      | FIRE    | Suspens | SL    | SVL   | BRAKE | UPS  | IOC   | ESTOP |

#### 17) Input Type OPB

- Input type setting of OPB-101 Board.
- “-“is Normal Open as A contact “B” is Normal Close as B contact.
- SBI3 is useful at the two doors.
- Control bit is as below.

| Parameter | 7             | 6             | 5      | 4        | 3             | 2      | 1      | 0             |
|-----------|---------------|---------------|--------|----------|---------------|--------|--------|---------------|
| SBI0      | 30%           | 45%           | FULL   | OVERLOAD | SAFETY        | PHOTO  | AIRCON | Not Specified |
| SBI1      | FIRE1         | FIRE2         | ATT    | AUP      | ADOWN         | ASTART | APASS  | ABZC          |
| SBI2      | OPEN          | CLOSE         | OPEN1  | OPEN2    | IND           | MH     | BBSW   | DSTP          |
| SBI3      | Not Specified | Not Specified | BPHOTO | BSAFETY  | Not Specified | BHOLD  | BCLOSE | BOPEN         |

|                                                                                   |                            |          |                  |
|-----------------------------------------------------------------------------------|----------------------------|----------|------------------|
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### 18) Speed Pattern

- Setting the RPM depends on speed.
- Input limit is 0~3600 rpm.

| Parameter | Display          | Initial value | Range    | Unit | Function                        |
|-----------|------------------|---------------|----------|------|---------------------------------|
| SPD0      | Zero Speed       | 0             | 0 ~ 3600 | rpm  | rpm for Zero speed.             |
| SPD1      | Creep Speed      | 40            |          | rpm  | rpm for Creep speed.            |
| SPD2      | Leveling Speed   | 100           |          | rpm  | rpm for rescue driving speed.   |
| SPD3      | Inspection Speed | 400           |          | rpm  | rpm for manual driving speed.   |
| SPD4      | Run Down Speed   | 800           |          | rpm  | rpm for run down speed          |
| SPD5      | Calibration      | 800           |          | rpm  | rpm for calibration.            |
| SPD6      | Short run Speed  | 1100          |          | rpm  | rpm for 1 floor driving speed   |
| SPD7      | Long run Speed   | 1450          |          | rpm  | rpm for 2 floors driving speed. |
| SPD8      | 3 run Speed      | 1450          | 0 ~ 3600 | rpm  | rpm for 3 floors driving speed  |
| SPD9      | 4 run Speed      | 1450          |          | rpm  | rpm for 4 floors driving speed  |
| SPD10     | 5 run Speed      | 1450          |          | rpm  | rpm for 5 floors driving speed  |

### 19) Fire Return Floor

- Setting the stopping floor when fire is identified.
- Setting the stopping floor when the fireman operates
- Setting the only 1stopping floor

### 20) Parking Floor

- As per PKS switch, setting the stopping when Parking driving.
- After service the registered call and park at the Paring Floor.

### 21) Homing Floor

- Setting the stopping floor when the generator drives at the Group mode.

### 22) Set Motor

- Input the parameter for Motor.

|                                                                                   |                            |          |                  |
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| Parameter     | Description                 | Initial value | Range      | Unit  | Effectiveness |
|---------------|-----------------------------|---------------|------------|-------|---------------|
| Motor RPM     | Input the Motor rpm.        | 1450          | 100 ~ 3600 | rpm   | 1             |
| Encoder Pulse | Input the Encoder Pulse no. | 1024          | 500 ~ 5000 | -     | 1             |
| Running Speed | Input the speed/min.        | 60            | 1 ~ 9999   | M/sec | Immediately   |

### 23) Set Time

- Setting the today's date and time.
- Use the ← and → Key for position moving, and use the ↑ **and** ↓ Key for changing no.
- Setting value is memorized when you press the [ENT] Key after changing

### 24) Set Address

- Setting the CP Address.
- It needs when you use the CRT.

|                                                                                   |                            |          |                  |
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### 3-2-4. OPT menu

#### 1) Inspection

##### ①. Direction LP

⇒ Setting the direction Lamp ON/OFF.

When set [OFF], direction lamp is not displayed even on operation in Manual Mode.

##### ②. Floor Display

⇒ Setting the floor display On/off.

When set [OFF], '- -' is displayed for Floor Display in Manual Mode.

##### ③. Door Open

⇒ setting the Door Open, or not

⇒ When set [Yes], Door is opened automatically if car is reached to Door Zone. If Manual UP, DOWN Button is activated, door is closed automatically.

⇒ When set [No], door is not opened in Manual Mode.

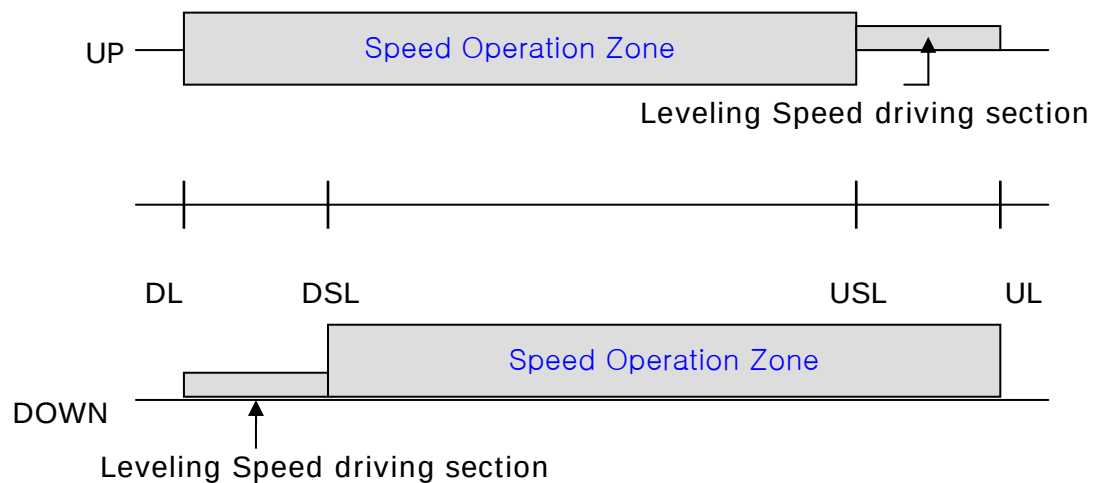
##### ④. Leveling Speed

⇒ Setting the driving part by leveling speed.

⇒ In Manual Mode, operation by inspection speed & Leveling speed is possible.

⇒ In case Drive Zone is set by UP:UL, DOWN:DL and Leveling speed is set by UP:USL, DOWN:DSL, operation speed is as follows.

|                                                                                   |                            |          |                  |
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#### ⑤. Drive Zone

⇒ Setting the manual driving Zone.

#### 2) Playing Call

- Set up ON/OFF for mischief call function.
- When load is 30%, will be recognized into playing call in case of registration over 4 floors.

#### 3) Reverse Call

- Set up ON/OFF for car call response function of reversed operating direction.

#### 4) Toggle Call

- Set up ON/OFF for car call toggle function.

#### 5) Abnormal

- Set up ON/OFF for disable function.

#### 6) Next Landing

- If you set [Yes], the door is opened at the nearest floor when the door is not opened.

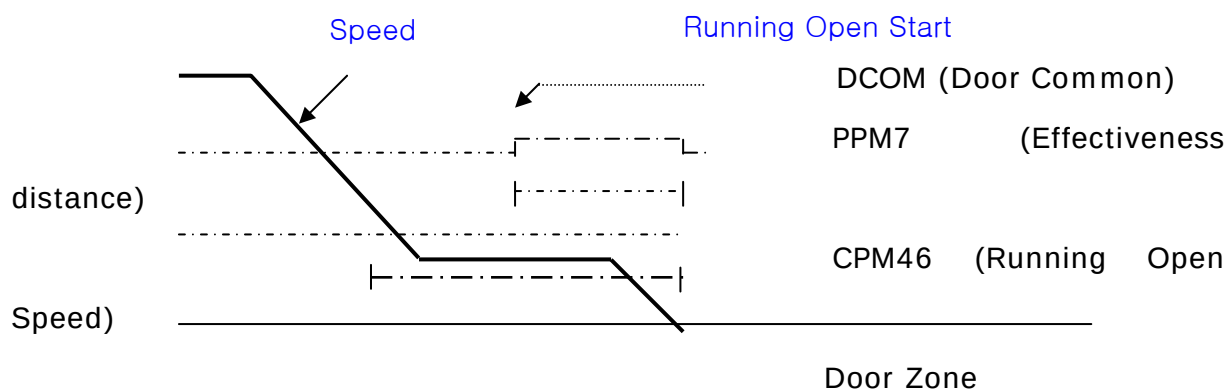
#### 7) Running Open

|                                                                                   |                            |          |                  |
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⇒Set up the [Yes] when the car runs open.

⇒Running Open Function need following action.

- ①. Running Open Option is to set [YES].
  - ②. Wiring is to be common for Car & Hall door by Door common Relay(DCOM)
- When the door is opened, as automatically operation is stopped, in case of Running Open, door must be common.
- ③. Set Running Open speed at CPM46. When running speed is less than CPM46, Running Open is possible.
  - ④. Set Running open point at PPM7. This can adjust Running Open distance.
  - ⑤. Running Open is possible only within Door Zone. When using Door Zone switch, Running Open distance is maximum 100mm and otherwise maximum 200mm



#### 8) Auto Leveling

- Set up ON/OFF for automatic level revision function.

#### 9) Brake Watch

- check the condition of Brake control signal.

#### 10) Defect Button

|                                                                                   |                            |          |                  |
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- Select the defect button.
- The defect button is divided from registration.

#### 11) Door Zone SW

- Set up ON/OFF for Door Zone switch function
- Have to input Door Zone Switch from the exterior for 'ON' function. .
- Will be considered as Door Zone, if either 'IUS' or 'IDS' is 'ON' function

#### 12) Load 45% SW

- In case of UPS operation for rescue operation, use this switch and decide the direction the direction of rescue operation..

#### 13) Photo Sensor

- Set up the photo sensor use or not.

#### 14) Separate OPB

- Set up to use for disabled car call from 21<sup>st</sup> floor to 40<sup>th</sup> floor hall lantern.

#### 15) Hall Lantern

- Select the display system as hall lantern.

#### 16) Repeat Door

- Set up ON/OFF of door open/close function for Auto repeat operation.

#### 17) OPP Auto Door

- Set up ON/OFF of Door open/close function for OPP operation..

#### 18) Fault Open

- Open waiting during Door fault.
- [No] is for Close waiting.

|                                                                                   |                            |          |                  |
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#### 19) Speed Upgrade

- Select when upgrade to one step higher speed for modernized construction.
- Stop switch hall
- Set up validity/invalidity of hall call function for stoppage of the service switch.

#### 20) Stop SW Hall

- As the service floor switch, if you want to be Hall Call service On/Off, set up to [Yes].
- If you set up [No], Service floor switch is On/Off for Car Call.

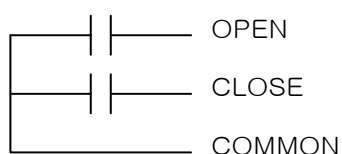
#### 21) Car Call First

- After Car Call service, Hall Call service.
- When there is hall call during car call service, the car doesn't stop.

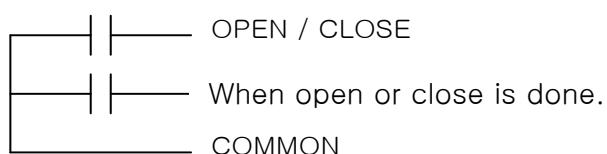
#### 22) Door One Signal

- Select the way of Door control signal output.
- Select [No], Open/Close signal is output by separate port.
- Select [Yes], Open/Close signal is output by one port.

It is always Open, or Close.



< In case of setting " No" >



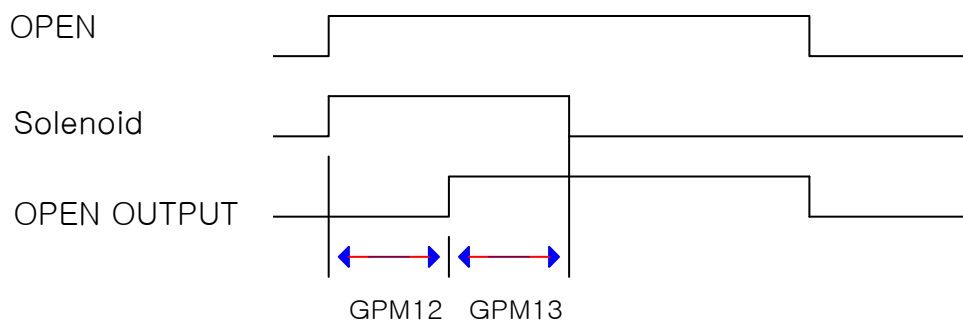
< In case of setting " Yes" >



|                                                                                   |                            |          |                  |
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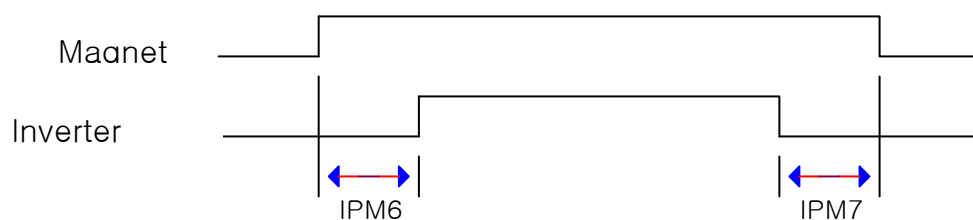
### 23) Door Solenoid

- When you use the Door Solenoid, select [Yes].
  - GPM12: Set up the On time of Solenoid before Door Open signal output.
  - GPM13: Set up the off pint of Solenoid after Door Open signal output.
- When you set up (0), Solenoid is On during the Door is open.

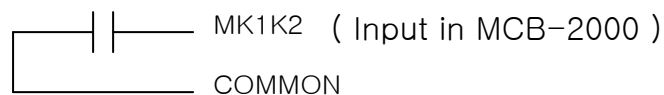


### 24) Magnet Control

- Use the control magnet between Inverter output and Motor, set up [Yes].
- IPM6: set up the time Magnet On before Inverter driving.
- IPM7: set up the time of Magnet Off after Inverter driving.



Stop the Inverter driving between Magnet control output signal and Monitoring input signal difference for 2 seconds.



### 25) Buzzer On/Off

- Buzzer On/Off when you operate the key of OPP-2000.

|                                                                                   |                            |          |                  |
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### 3-2-5. PROG menu

#### 1) Parking

- Parking function ON/OFF is possible by program setting.
- Maximum 6 program is possible and ON/OFF need separate program.
- Separate allocation for Parking Floor is possible for each program.
- Parking is the function which stops operation on at allocated floor.
- ON/OFF condition of parking function by program setting can be checked at RSV of variable Menu. When it is On, PAK is displayed in Gray.

#### 2) Schedule

- Schedule operation ON/OFF is possible by program setting.
- Maximum program can be by 8, and On/Off Program is separate
- Set up for Stand By floor at each Program
- Time change of Schedule operation can be set on GPM4.
- Schedule operation is function for Stand By with door closing at allocated floor during the setting time at GPM4 in case there is no Car/Hall call.
- Schedule operation is function for Stand By with door closing at allocated floor during the setting time at GPM4 in case there is no Car/Hall call.

#### 3) Each Floor Stop (each floor stop)

- Each Floor stop On/OFF.
- Program can be by 6, and On/Off Program is separate
- Each Floor stop operation is the function which stops operation at each floor until reaching at destination floor regardless of call registration. During the Each Floor stop operation, close button does not work.

|                                                                                   |                            |          |                  |
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Program 1 :

From :

To :

Time :

Set :                      FLR :

SET
ESC
REMOVE
PAK

### < The way of Program >

- For example, when you are Off the parking every 8:00 AM, and On at 10:00 PM, it is as below.
- When you select the parking on the program menu and the press the [ENT] Key, it show the display of program no.
- You can see the program window after press the P1 and select the [ENT] Key
  - 1) From: select the Everyday. The day is changed whenever you press the [ENT] Key If it is Everyday, no need to set up To:
  - 2) Time: set up Am08:00, whenever you press the [ENT] Key, the time is changed.
  - 3) Set: Set up setting /Release of parking
  - 4) FLR: Set up for parking floor.
  - 5) SET. Memory of setting value
  - 6) Set up the same way at 10:00PM on P2, and setting up Set : ON.

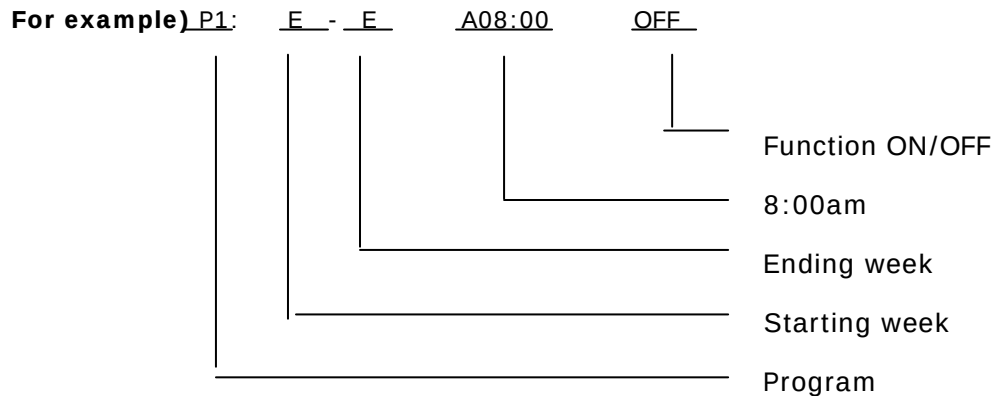
### < Program Window Description >

As per P1, parking cancel every 8:00am, and parking On every 10:00pm

- 1) From: setup starting week of program.
- 2) To : setup ending week of program.
- 3) Time: setup starting time of program.
- 4) Set: setup On/Off function.
- 5) FLR: setup for parking floor.
- 6) SET: Memory setting value.
- 7) ESC: cancel of setting value(return to prior value).
- 8) REMOVE: Elimination of program.
- 9) PAK, SCH, EFS indicate the status of execution by programming setting.

|                                                                                   |                            |          |                  |
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### < Program registration window >



### 3-2-6. DISP menu

#### 1) Floor Display

To be changed the floor display by ← and → Key, after set cursor on the designated floor and press [ENT] Key.

##### 1) Edit: Indicate character of floor display.

can display maximum 4 digits, alphabet and special character.

setup by number key or ↑ **and** ↓ Key, but setup the English and special character by ↑ **and** ↓ Key.

##### 2) Delete: A relevant floor indication is deleted, and a character of the last floor becomes a copy as it is.

##### 3) Insert : Add a floor indication character

##### 4) Default: Initializing Floor Display Character to 1~40 Floor.

#### 2) Indicator Lamp

- There are possible output of 8 program on the Indicator.
- You can select and output one of them for each lamp.

| No | Name         | Function                                                                                                          |
|----|--------------|-------------------------------------------------------------------------------------------------------------------|
| 1  | UP           | Lamp for UP driving direction.                                                                                    |
| 2  | Down         | Lamp for DOWN driving direction.                                                                                  |
| 3  | Full         | Light On when it is full.                                                                                         |
| 4  | Inspection   | Light On when it is inspection.                                                                                   |
| 5  | Auto         | Light On when it is Auto mode                                                                                     |
| 6  | Overload     | Light On when it is Overload                                                                                      |
| 7  | Attendant    | Light On when it is Attendant driving.                                                                            |
| 8  | Independent  | Light On when it is Independent driving.                                                                          |
| 9  | Fireman      | Light On when it is Fireman driving.                                                                              |
| 10 | Parking      | Light On when it is parking.                                                                                      |
| 11 | Seismic      | Light On when it is Seismic.                                                                                      |
| 12 | UPS          | Light On when it is Light off.                                                                                    |
| 13 | Fire         | Light On when it is Fire.                                                                                         |
| 14 | Controlling  | Light On when it is Seismic, UPS, Fire                                                                            |
| 15 | Get Off      | Light On when it is Seismic, UPS, Fire.                                                                           |
| 16 | Occupied     | Suspense                                                                                                          |
| 17 | Moving House | Light On when it is Moving.                                                                                       |
| 18 | Operation    | Light On when it is Auto driving mode, but not fault.                                                             |
| 19 | Fault        | Light On when it is Fault.                                                                                        |
| 20 | EXT1         | Suspense                                                                                                          |
| 21 | Generator    | Light On when it is Generator driving.                                                                            |
| 22 | Control End  | Light On when Seismic, UPS, Fire Parking and generator are finished, or auto mode is impossible because of fault. |
| 23 | EXT4         | Light On when the car is driving                                                                                  |
| 24 | EXT5         | Light On while the car stops.                                                                                     |

### 3-2-7. VER menu

- Indicate Micro Processor for each board.

| No | Name     | Description                                      |
|----|----------|--------------------------------------------------|
| 1  | CPU-2000 | It is the Version of CPU program.                |
| 2  | OPP-2000 | It is the Version of OPP-2000 program.           |
| 3  | GCU-2000 | It is the Version of Group Controller program.   |
| 4  | ESI-2000 | It is the Version of ESI(EPM,EPS) program..      |
| 5  | OPB-2000 | It is the Version of OPB program on the Car top. |
| 6  | PSD-2000 | It is the Version of main program.               |
| 7  | INV-2000 | It is the Version of Inverter program.           |

|                                                                                   |                            |          |                  |
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### 3-2-8. PWORD menu

#### 1) Master Password

##### ① Pword ON/OFF

- Setup [Yes], Password function is available.
- Setup [No], Password function is unavailable when you contact to OPP-2000.
- When you choose 'Yes' after pressing [ENT] Key, Yes/No is changed.

##### ② Pword Check

- Select [ALL]: Check up every Off/On.
- Select 'OVT': When source of electricity 'OFF' and elapse 20minutes, check up in case of 'ON'.
- When you choose 'Yes' after pressing [ENT] Key, ALL/OVT is changed.

##### ③ Set General Pword

- When OPP is connected, password needs to enter into the main screen.
- It is possible to use the Master Password instead of General Password.
- Basic Password is 1234.
- Password is number and can setup by 8 numbers.

##### ④ Set Keep Pword

- It is password when you enter to KEEP on the CATEGORY.
- It is possible to use the Master Password instead of General Password.
- Basic Password is 1234.
- Password is number and can setup by 8 numbers.

##### ⑤ Set Master Pword

- It is password when you enter to PWORD on the CATEGORY.
- You can use the Master Password at the General Password or Keep Password.
- Basic Password is 1234.
- Password is number and can setup by 8 numbers.

|                                                                                   |                            |          |                  |
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## ⑥ Factory Setup

- All Data is set to the factory setting value
- Calibration Data, Faulty and warranty Report data are set.

## 2) Keep Password

### ① Set Keep Pword

- ⇒ Password to order when enters KEEP during CATEGORY.
- ⇒ Instead of KEEP Password, Master Password is also accessible.  
Default Password is 1234.
- ⇒ Password is numerical, and it is possible by figure largest 8.

## 3) General Password

### ① Pword ON/OFF

- ⇒ A Password function works with effectiveness if you set up as [Yes].  
When set by [No], OPP-2000 does not Ask Password.
- ⇒ If you presses [ENT] Key and selects Yes, it is turned over Yes/No.

### ② Pword Check

- ⇒ Incase of setting [All], when every Power OFF → ON, it ask Password.
- ⇒ Incase of setting [OVT], once Password is set, If Power is on again in 20 min, after Power off. It dose not ask Password.
- ⇒ If you presses [ENT] Key and selects Yes, it is turned over ALL/OVT.

### ③ Set General Pword

- ⇒ This Password is asked before entering Basic Display when connection OPP.
- ⇒ Instead of General Password, Master Password is also accessible.  
Basic Password rises in 1234.
- ⇒ Password is numerical, and it is possible by figure largest 8.

|                                                                                   |                            |          |                  |
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### 3-2-9. KEEP menu

#### 1) Erase Repot ALL

- Faulty and Warranty data are all erased
- Once it is erased, it doesn't recovery.

#### 2) Report Write

- As it sets [YES], the Faulty and Warranty is recorded.
- As it sets [No], Faulty and Warranty is recorded.

#### 3) Set System Down Time

- It programs the system inspection time.
- Direction move is ← and → Key, number change is ↑ **and** ↓ Key.
- It is memorized when you press the [ENT] Key.
- The car doesn't move anymore when it is setting time.
- For re-working, you have to remove the program with **REMOVE**.

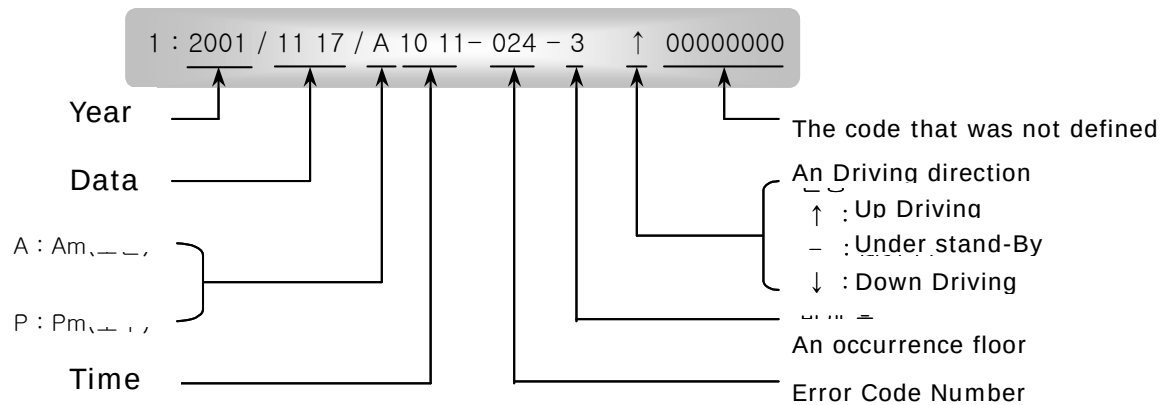


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## 4. Error Code

### 4-1. Error Code Type

Type : Year – date – Time – Error Code – Fault floor – Driving direction –  
Others No use the 'Others' right now.



### 4-2. Controller Error Code, Table1.

| CODE | Display          | Description                                                                                                                                                                                                 |
|------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Non Off Door SW  | <ul style="list-style-type: none"> <li>- Door switch is not OFF.</li> <li>- Check whether Door is Common.</li> <li>- Check the Car Door or Hall Door Switch.</li> </ul>                                     |
| 2    | Non Off DCL      | <ul style="list-style-type: none"> <li>- DCL is not OFF.</li> <li>- Check the Door Close Limit.</li> </ul>                                                                                                  |
| 3    | No Open Door     | <ul style="list-style-type: none"> <li>- Door is not open.</li> <li>- Check the both of DOL and DCL are ON.</li> <li>- Check the Door.</li> </ul>                                                           |
| 4    | No Input Door SW | <ul style="list-style-type: none"> <li>- Door switch is not input.</li> <li>- Check the Door or Door switch.</li> </ul>                                                                                     |
| 5    | No Input DCL     | <ul style="list-style-type: none"> <li>- DCL isn't input.</li> <li>- Check the Door Close Limit.</li> </ul>                                                                                                 |
| 6    | No Close Door    | <ul style="list-style-type: none"> <li>- Door isn't closed.</li> <li>- Check the Door.</li> </ul>                                                                                                           |
| 7    | UL On            | <ul style="list-style-type: none"> <li>- UL is working.</li> <li>- UL doesn't work at the top floor.</li> <li>- Check the position or switch of UL.</li> </ul>                                              |
| 8    | SUSL On          | <ul style="list-style-type: none"> <li>- Before deceleration at the top floor, SUSL works.</li> <li>- Check the setting time of IPM5.</li> <li>- Fix the USL position or setting timing of IPM5.</li> </ul> |
| 9    | USL ON           | <ul style="list-style-type: none"> <li>- USL works.</li> <li>- Fix the USL position,</li> </ul>                                                                                                             |
| 10   | HUSL ON          | <ul style="list-style-type: none"> <li>- HUSL works.</li> <li>- Fix the HUSL position.</li> </ul>                                                                                                           |
| 11   | VHUSL ON         | <ul style="list-style-type: none"> <li>- VHUSL works.</li> <li>- Fix the VHUSL position.</li> </ul>                                                                                                         |
| 12   | MHUSL ON         | <ul style="list-style-type: none"> <li>- MHUSL works.</li> <li>- Fix the MHUSL direction.</li> </ul>                                                                                                        |
| 13   | DL ON            | <ul style="list-style-type: none"> <li>- DL works.</li> <li>- DL shouldn't work over bottom floor level.</li> <li>- Check the DL position or switch.</li> </ul>                                             |
| 14   | SDSL ON          | <ul style="list-style-type: none"> <li>- SDSL works before deceleration at the bottom floor.</li> <li>- Check the setting time of IPM4.</li> <li>- Fix the DSL position or setting time of IPM4.</li> </ul> |
| 15   | DSL ON           | <ul style="list-style-type: none"> <li>- DSL works.</li> <li>- Fix the DSL position.</li> </ul>                                                                                                             |

| CODE | Display          | Description                                                                                                                |
|------|------------------|----------------------------------------------------------------------------------------------------------------------------|
| 16   | HDSL ON          | - HDSL works.<br>- Fix the HDSL position.                                                                                  |
| 17   | VHDSL ON         | - VHDSL works.<br>- Fix the VHDSL position                                                                                 |
| 18   | MHDSL ON         | - MHDSL works.<br>- Fix the MHDSL position                                                                                 |
| 19   | Overrun          | - Running time of one floor is over.<br>- Check the GPM1 time.                                                             |
| 20   | Overrun Level    | - Didn't start.                                                                                                            |
| 21   | Over Stop        | - It passed and stopped the Level.<br>- Extend the deceleration at PPM.                                                    |
| 22   | Small Over       | - It passed a little bit and stopped the Level.<br>- Extend the deceleration at PPM.                                       |
| 23   | Low Creep Speed  | - It didn't reach at the Level.<br>- When the Creep is long, reduce the distance of deceleration.<br>- Up the Creep speed. |
| 24   | Inverter Fault   | - Inverter Error happens.<br>- Check the inverter.                                                                         |
| 25   | No Input Pulse   | - Encoder Pulse is not input.<br>- Check the Encoder connection.                                                           |
| 26   | Floor Error PE   | - Floor position is wrong.<br>- Check the Encoder or IUS, IDS.                                                             |
| 27   | Floor Error PEL  | - Floor position is wrong.<br>- Check the Encoder or IUS, IDS.                                                             |
| 28   | Floor Error PUOE |                                                                                                                            |
| 29   | Floor Error PDOE |                                                                                                                            |
| 30   | Floor Error PUFE |                                                                                                                            |
| 31   | Floor Error PDFE |                                                                                                                            |
| 32   | Floor Error FC   |                                                                                                                            |
| 33   | No Input USL     | - USL is not input at the top floor.<br>- Check the USL switch.<br>- Check the Encoder or IUS, IDS.                        |
| 34   | No Input HUSL    | - No HUSL input.<br>- Check the HUSL switch.                                                                               |
| 35   | No Input VHUSL   | - No VHUSL Input.<br>- Check the VHUSL switch.                                                                             |
| 36   | No Input MHUSL   | - No MHUSL input.<br>- Check the MHUSL switch.                                                                             |
| 37   | No Input DSL     | - DSL isn't input at the bottom floor.<br>- Check the DSL switch.<br>- Check the Encoder or IUS, IDS.                      |
| 38   | No Input HDSL    | - No HDSL input.<br>- Check the HDSL switch.                                                                               |

| 39   | No Input VHDSL    | - No VHDSL input.<br>- Check the VHDSL switch.                                                                                                                 |
|------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 40   | No Input MHDSL    | - No MDSL.<br>- Check the MDUSL switch.                                                                                                                        |
| CODE | Display           | Description                                                                                                                                                    |
| 41   | No Input IUS      | - No IUS Input.<br>- Check the Encoder and IUS.                                                                                                                |
| 42   | No Input IDS      | - No IDS Input.<br>- Check the Encoder and IDS.                                                                                                                |
| 43   | No Input IUS/IDS  | - Both of IUS and IDS no input.<br>- Check the Encoder and IUS,IDS                                                                                             |
| 44   | No Input DZONE    | - No Door Zone switch input.<br>- When you don't use the Door Zone switch, set the Door Zone SW [No] at the [OPT].<br>- Check the Door Zone switch.            |
| 45   | Not Off IUS       | - IUS isn't OFF.<br>- Check the IUS.                                                                                                                           |
| 46   | Not Off IDS       | - IDS isn't OFF<br>- Check the IDS.                                                                                                                            |
| 47   | Not Off IUS/IDS   | - Both of IUS and IDS aren't OFF<br>- Check both of IUS and IDS.                                                                                               |
| 48   | Not Off DZONE     | - Door Zone Switch isn't OFF<br>- Check the DZONE Switch.                                                                                                      |
| 49   | Failed leveling   | - Can't find out the Level.<br>- Check the IUS and IDS.                                                                                                        |
| 50   | Door Opened       | - The door is open during driving.<br>- Check the Door.                                                                                                        |
| 51   | IUS/IDS Abnormal  | - There is no point IUS and IDS ON at the same time.<br>- Do the calibration after reducing the distance between IUS and IDS or reducing the Calibration speed |
| 52   | IUS/IDS Exchange  | - The position of IUS and IDS are changed.<br>- Change the position of IUS and IDS.                                                                            |
| 53   | PA/PB Exchange    | - The position of PA and PB are changed.<br>- Change the PA and PB.                                                                                            |
| 54   | Floor Abnormal    | - There are too many floors.<br>- The available running floor is maximum 40 stories.<br>- Check the On/Off condition of IUS and IDS.                           |
| 55   | Calibration Fail  | - Calibration fail.                                                                                                                                            |
| 56   | Communication OPB | - OPB communication error<br>- Check the OPB communication LINE.                                                                                               |
| 57   | Communication IND | - IND communication error.<br>- Check the IND communication LINE.                                                                                              |
| 58   | Communication GCU | - GCU communication error.<br>- Check the GCU communication LINE.                                                                                              |

| 59   | USL On Error     | <ul style="list-style-type: none"> <li>- USL is On out of top floor</li> <li>- Check the USL Switch.</li> <li>- Check the Encoder or IUS, IDS.</li> </ul>                                                                                                                                                                |
|------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 60   | DSL On Error     | <ul style="list-style-type: none"> <li>- DSL is On out of the bottom floor.</li> <li>- Check the DSL Switch</li> <li>- Check the Encoder or IUS, IDS.</li> </ul>                                                                                                                                                         |
| CODE | Display          | Description                                                                                                                                                                                                                                                                                                              |
| 61   | Roll Back        | <ul style="list-style-type: none"> <li>- Roll Back is happened.</li> <li>- The running direction and Motor control direction is different.</li> <li>- Check the Encoder connection and Inverter input connection.</li> <li>- When the roll back is happened during connection is correct, check the Inverter.</li> </ul> |
| 62   | UPS Input On     | - UPS driving mode                                                                                                                                                                                                                                                                                                       |
| 63   | Fire Input On    | - Fire signal is input.                                                                                                                                                                                                                                                                                                  |
| 64   | Generator On     | - Generator driving mode.                                                                                                                                                                                                                                                                                                |
| 65   | Seismic Very Low | - Seismic very low signal input.                                                                                                                                                                                                                                                                                         |
| 66   | Seismic Low      | - Seismic very low signal input                                                                                                                                                                                                                                                                                          |
| 67   | Seismic High     | - Seismic very low signal input.                                                                                                                                                                                                                                                                                         |
| 68   | Watchdog GCU     | <ul style="list-style-type: none"> <li>- GCU and communication is abnormal.</li> <li>- Check the GCU and communication Line.</li> </ul>                                                                                                                                                                                  |
| 69   | Watchdog ESI     | <ul style="list-style-type: none"> <li>- ESI and communication is abnormal.</li> <li>- Change the Main Board.</li> </ul>                                                                                                                                                                                                 |
| 70   | UL On Error      | <ul style="list-style-type: none"> <li>- UL is On of Top floor.</li> <li>- Check the UL switch.</li> </ul>                                                                                                                                                                                                               |
| 71   | DL On Error      | <ul style="list-style-type: none"> <li>- DL is On out of bottom floor</li> <li>- Check the DL switch.</li> </ul>                                                                                                                                                                                                         |
| 72   | Door Drive Error | <ul style="list-style-type: none"> <li>- DOOR Drive is error</li> <li>- Check the Door Drive.</li> </ul>                                                                                                                                                                                                                 |
| 73   | Power Good Error | - Power part is error.                                                                                                                                                                                                                                                                                                   |
| 74   | Power Off        | <ul style="list-style-type: none"> <li>- Power is off during driving.</li> <li>- When you do the power off and on, the car does to the bottom floor and is normal.</li> </ul>                                                                                                                                            |
| 75   | No Open Brake    | <ul style="list-style-type: none"> <li>- No open Brake.</li> <li>- IF you don't use the Brake Monitoring function, set up the brake watch [No] at the [OPT].</li> <li>- Check the Brake and Brake Monitoring input.</li> </ul>                                                                                           |
| 76   | No Close Brake   | <ul style="list-style-type: none"> <li>- No Brake close.</li> <li>- IF you don't use the Brake Monitoring function, set up the brake watch [No] at the [OPT].</li> <li>- Check the Brake and Brake Monitoring input.</li> </ul>                                                                                          |

| 77   | Oil Temp Low       | <ul style="list-style-type: none"> <li>- Low temperature sensor is working at the hydraulic.</li> <li>- Automatically working during low temperature sensor.</li> </ul> |
|------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 78   | Oil Temp High      | <ul style="list-style-type: none"> <li>- High temperature sensor is working at the hydraulic.</li> <li>- Return to the bottom floor and no more running.</li> </ul>     |
| 79   | Non Off A door SW  | - A Door switch is not Off at the double doors.                                                                                                                         |
| 80   | Non Off DCLA       | - A Door Close Limit is not OFF at the double doors                                                                                                                     |
| 81   | No Open A door     | - A Door is not open at the double doors.                                                                                                                               |
| 82   | No Input A door SW | - A Door switch is not input at the double doors .                                                                                                                      |
| CODE | Display            | Description                                                                                                                                                             |
| 83   | No Input DCLA      | - A Door Close Limit is not input at the double doors.                                                                                                                  |
| 84   | No Close A door    | - A Door sin not close at the double doors.                                                                                                                             |
| 85   | Non Off B door SW  | - B Door switch is not Off at the double doors.                                                                                                                         |
| 86   | Non Off DCLB       | - B Door Close Limit is not OFF at the double doors.                                                                                                                    |
| 87   | No Open B door     | - B Door in not open at the double doors.                                                                                                                               |
| 88   | No Input B door SW | - B Door switch is not input at the double doors.                                                                                                                       |
| 89   | No Input DCLB      | - B Door Close Limit is not input at the double doors                                                                                                                   |
| 90   | No Close B door    | - B Door is not close at the double doors.                                                                                                                              |
| 91   | System Down        | <ul style="list-style-type: none"> <li>- It is time to system check.</li> <li>- After checking the system and cancel the setting.</li> </ul>                            |
| 92   | Magnet Error       | <ul style="list-style-type: none"> <li>- Magnet doesn't work correct.</li> <li>- Check the Magnet between Inverter and Motor.</li> </ul>                                |
| 93   | Watchdog COMM      | <ul style="list-style-type: none"> <li>- Communication Error between MCU and Inverter</li> <li>- Replace the Main Control Board</li> </ul>                              |
| 94   | Emergency Stop     | - E-Stop Switch have been operated                                                                                                                                      |
| 95   | Input Signal Error | - Fault Input Signal is coming. Please check whether certain Input is activating or not.                                                                                |
| 96   | 'Rope Brake ON'    | Rope Brake is activated                                                                                                                                                 |
| 97   | 'Rope Brake IN'    | Input Rope Brake Operation Signal                                                                                                                                       |
| 98   | 'Rope Brake Error' | Floor Limit Error Happened                                                                                                                                              |
| 99   | 'Fault 129'        | No Input DZONE SW                                                                                                                                                       |
| 100  | 'Fault 130'        | Non Off DZONE SW                                                                                                                                                        |
| 251  | Hall Call Up       | - It is Hall Call Up button trouble.                                                                                                                                    |

|     |                   |                                                 |
|-----|-------------------|-------------------------------------------------|
| 252 | Hall Call Down    | - It is Hall Call Down button trouble.          |
| 254 | Ab Hall Call Up   | - It is disabled Hall Call Up button trouble.   |
| 255 | Ab Hall Call Down | - It is disabled Hall Call Down button trouble. |

### 4-3. Inverter Error Code, Table 2

| CODE | Display        | Description                                                                                                                                                                                                                                                                                                                                                  |
|------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 101  | IPM Fault      | <ul style="list-style-type: none"> <li>- IPM overheat or ZPM over current.</li> <li>- Overload or too short Acc/Deceleration time</li> <li>- Improper Motor Tuning or Time Constant of Rotor, CPM19.</li> </ul>                                                                                                                                              |
| 102  | Over Voltage   | <ul style="list-style-type: none"> <li>- Over voltage alarm of DC bus voltage.</li> <li>- In case of IPM Error on installation, check Input Error for peripeal devices such as Encoder, Brake timing, Parameter.</li> </ul> <p>Consistent operation without troubleshoot can make IPM damaged.</p> <p>⇒ Deceleration time is short</p> <p>⇒ Over load</p>    |
| 103  | Under Voltage  | <ul style="list-style-type: none"> <li>- Low voltage alarm of DC bus voltage.</li> <li>- In the case of 3phase input is 190V, this problem happens when the bus voltage over 380V, In the case of 3phase input is 400V, this problem happens when the bus voltage is over 800V.</li> </ul> <p>⇒ Too low the 3phase input voltage</p>                         |
| CODE | Display        | Description                                                                                                                                                                                                                                                                                                                                                  |
| 104  | Motor Overload | <ul style="list-style-type: none"> <li>- Motor overload alarm.</li> <li>- It happens when the 5Motor load current comes 200% for 1 second.</li> </ul> <p>⇒ Check the Overload</p> <p>⇒ Check whether the load is locked by brake or others.</p> <p>⇒ Motor tuning is incorrect. Especially the setting of rotor time constant(Tr) of CPM19 is incorrect.</p> |

|     |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 105 | Over Speed         | <ul style="list-style-type: none"> <li>- Alarm of over speed.</li> <li>- If the motor rotor is over 110% of regular speed, the problem is happened. For example, when the motor is 4phase with 50HZ, regular speed is 1500 rpm. However if the motor rotor is over 1650 rpm, this alarm comes out.</li> <li>⇒ Speed can't control.</li> <li>The reason would be the encoder signal problem.</li> </ul>                                                                                                                                                                                                                                                                                                                                               |
| 106 | Excess Speed Error | <ul style="list-style-type: none"> <li>- This problem happens when the difference of motor order rotor and real rotor is over 20% of regular speed. For example, the regular speed is 1500rpm when the motor is 4phase with 50HZ, but in spite of it orders 1500 rpm, if it rotates under 1200 rpm, the alarm comes out.</li> <li>⇒ The alarm comes out during deceleration by manual operation, the reason is it gives the deceleration order suddenly, so make the deceleration order slowly.</li> <li>⇒ Check whether the torque lacks because the load is over.</li> <li>⇒ Check whether the ac/deceleration time is too short.</li> <li>⇒ It happens when the input is lower than real value of no load current at the motor tuning.</li> </ul> |
| 107 | DSP Fault1         | <ul style="list-style-type: none"> <li>- Alarm for DSP chip error.</li> <li>- No alarm at the normal condition.</li> <li>However, if there is a lot of noise, the alarm can work.</li> <li>⇒ Check the surrounding noise.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 108 | DSP Fault2         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 109 | Motor Parameter    | <ul style="list-style-type: none"> <li>- Motor parameter error. For example, when you input more value of no load current than load current, or as per auto tuning, you can't decide the motor parameter..</li> <li>After input the correct motor data, reset or in out power again.</li> <li>⇒ Input the correct parameter.</li> <li>⇒ When the auto tuning is fail, try again by manual operation.</li> </ul>                                                                                                                                                                                                                                                                                                                                      |



| CODE | Display        | Description                                                                                                                                                                                                                                |
|------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 110  | Low Power      | <p>- This problem happens when the current is not increased after input the power at the DC bus.</p> <p>This problem also happens when the power input to the control plate.</p> <p>⇒ Check the R, S, T 3phase connection and voltage.</p> |
| 111  | Encoder Input  | <p>- When the motor rotor direction and encoder feed back direction is different for 2 seconds, this problem happens.</p> <p>⇒ Wrong connection at the encoder A and B</p>                                                                 |
| 112  | Over Heat      | <p>- The temperature of protecting heating plate of electronic part is over 85 °C.</p> <p>⇒ Too much driving</p> <p>⇒ Cool fan fault</p>                                                                                                   |
| 113  | Watchdog INV   | <p>-This problem happens when the MCU unit fault or Inverter communication error.</p> <p>⇒ Replace the main board</p>                                                                                                                      |
| 114  | CT Input       | <p>⇒ Current sensor is not contacted</p> <p>⇒ Check the contact status</p>                                                                                                                                                                 |
| 115  | Over Current U | <p>⇒ U phase over current</p> <p>⇒ Check the capacity of Inverter</p> <p>⇒ Check the capacity of Counter Weight</p> <p>⇒ Check the Motor Input parameter</p> <p>⇒ Check whether Acc/Deceleration time is too short</p>                     |
| 116  | Over Current V | <p>⇒ V phase over current</p> <p>⇒ Check the capacity of Inverter</p> <p>⇒ Check the capacity of Counter Weight</p> <p>⇒ Check the Motor Input parameter</p> <p>⇒ Check whether Acc/Deceleration time is too short</p>                     |
| 117  | Over Current W | <p>⇒ W phase over current</p> <p>⇒ Check the capacity of Inverter</p> <p>⇒ Check the capacity of Counter Weight</p> <p>⇒ Check the Motor Input parameter</p> <p>⇒ Check whether Acc/Deceleration time is too short</p>                     |

|                                                                                   |                            |          |                  |
|-----------------------------------------------------------------------------------|----------------------------|----------|------------------|
|  | <b>MMR Elevator</b>        | Rev. No. | 0                |
|                                                                                   |                            | Doc. No. | DL<br>EL) 06-007 |
| <b>FOD</b><br>Field Industrial Engineering                                        | <b>Installation Manual</b> | Page     | 82/107           |
|                                                                                   |                            | Date     | June. 2006       |

## **5. Appendix (Applies to OPP Version 3.0 above or equal to)**

### **5-1. Adjustment of LCD bright**

- Press Fn + ☀ to make it bright.
- Press Fn + ☿ to make it dark.

### **5-2. Function of shortcut key**

- 8 Key + 0: MODE
- 8 Key + 1: OPP AUTO
- 8 Key + 2: Set Motor
- 8 Key + 3: PPM
- 8 Key + 4: CPM
- 8 Key + 5: SPM
- 8 Key + 6: Warning Report
- 8 Key + 7: Fault Report

### **5-3. To move between menus**

- Current menu window is memorized when Fn + Mem is pressed.  
Max. 5 menu windows can be memorized.  
If 6<sup>th</sup> window is memorized, the 1<sup>st</sup> window memorized is lost.
- With menu windows memorized, if Fn + ► or Fn + ◀ is pressed,  
then next or previous menu window appears respectively.

### **5-4. Data Uploading & Data Downloading**

- If enter 'KEEP' menu, the following 4, 5 items are displayed.
  1. Erase Report All
  2. Report Write [Yes]
  3. Set System Down Time >
  4. Data Uploading >
  5. Data Downloading >
- If you select 'Data Uploading', new window appears as follows.

|                                                                                   |                            |          |                  |
|-----------------------------------------------------------------------------------|----------------------------|----------|------------------|
|  | <b>MMR Elevator</b>        | Rev. No. | 0                |
|                                                                                   |                            | Doc. No. | DL<br>EL) 06-007 |
| <b>FOD</b><br>Field Industrial Engineering                                        | <b>Installation Manual</b> | Page     | 83/107           |
|                                                                                   |                            | Date     | June. 2006       |

Select NO:

1. M1: 2004/1115/A1126 <- Means time and date of uploading from CP
2. M2: ---- ---- ----- <- Means uploaded data empty
3. M3: ---- ---- ----- <- Means uploaded data empty

-. If you press 'ENT' key at cursor, popup window appears as follows.

1. Uploading
2. Delete

. When there is previously uploaded data like M1, if you select '1.Uploading' it asks whether it will overwrite previous data. If you select 'Yes(ENT)' new data from CP will be read and saved.

. When there is no previous uploaded data like M2, if you select

1. Uploading', then new data is read immediately from CP

.When uploading finished, press [ESC] Key or [ENT] Key to complete uploading.

-. If you select 'Data downloading', new window appears as follows.

Select NO:

1. M1: 2004/1115/A1126 <- Means time and date of uploading from CP
2. M2: ---- ---- ----- <- Means uploaded data empty
3. M3: ---- ---- ----- <- Means uploaded data empty

-. If you press 'ENT' key at cursor, popup window appears as follows.

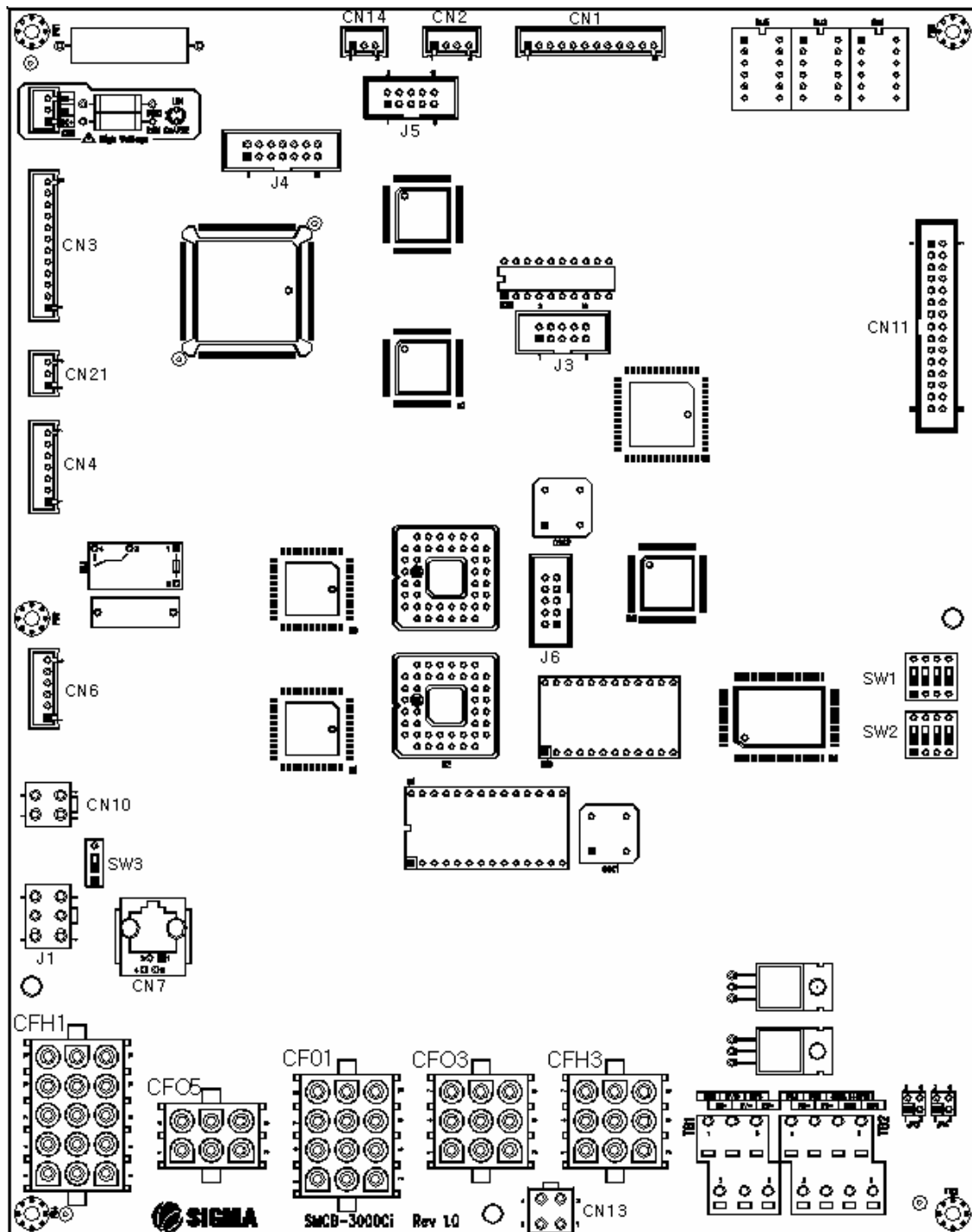
1. Downloading
2. Delete

. When there is previously uploaded data like M1, if you select '1.Downloading' it immediately downloads selected data to CP.

. Since calibration data is not copied, after downloading you must commence new calibration.

### 3. SMCB-3000Ci (Main Control Board)

#### 1. SMCB-3000Ci (Configuration)



### 2. Connector explanation

#### 1) CN1 (power supply part)

| Pin no | Pin name | I/O | Function                   |
|--------|----------|-----|----------------------------|
| 1      | C+12     | -   | Communication power +12V   |
| 2      | CGND     | -   | Communication power Ground |
| 3      | C-12     | -   | Communication power -12V   |
| 4      | SCOM     | -   | Switch input power Ground  |
| 5      | S+24V    | -   | Switch input power 24V     |
| 6      | GND      | -   | VDD,VCC,VSS Power Ground   |
| 7      | VSS      | -   | Power for OP Amp-15V       |
| 8      | GND      | -   | VDD,VCC,VSS Power Ground   |
| 9      | VCC      | -   | Power for OP Amp+15V       |
| 10     | GND      | -   | VDD,VCC,VSS Power Ground   |
| 11     | VDD      | -   | Digital Power 5V           |
| 12     | VIN      | -   | Power Good Signal          |

#### 2) CN2

| Pin no. | Pin name | I/O | Function                                        |
|---------|----------|-----|-------------------------------------------------|
| 1       | VDC+     | I   | Inverter electric device DC bus voltage sensor+ |
| 2       | VDC-     | I   | Inverter electronic part DC bus voltage sensor- |
| 3       | NC       | -   |                                                 |
| 4       | REGEN    | O   | Regeneration drive output                       |

#### 3) CN3

| Pin no. | Pin name | I/O | Function                         |
|---------|----------|-----|----------------------------------|
| 1       | GND      | -   | VDD,VCC,VSS power Ground         |
| 2       | VDD      | -   | Digital power 5V                 |
| 3       | FW       | I   | Inverter IPM UPPER W fault       |
| 4       | WP       | O   | Inverter W UPPER running signal  |
| 5       | FV       | I   | IPM UPPER V fault                |
| 6       | VP       | O   | Inverter V UPPER working signal  |
| 7       | FU       | I   | Inverter IPM UPPER U phase fault |
| 8       | UP       | O   | Inverter U UPPER working signal  |
| 9       | FN       | I   | Inverter IPM LOWER LEG fault     |
| 10      | UN       | O   | Inverter U LOWER working signal  |
| 11      | VN       | O   | Inverter V LOWER working signal  |
| 12      | WN       | O   | Inverter W LOWER working signal  |

### 4) CN4

| Pin no | Pin name | I/O | Function                            |
|--------|----------|-----|-------------------------------------|
| 1      | VCC      | -   | OP Amp power +15V                   |
| 2      | VSS      | -   | OP Amp power -15V                   |
| 3      | HALLU    | I   | Inverter electronic current check U |
| 4      | GND      | -   | VDD,VCC,VSS power Ground            |
| 5      | HALLV    | I   | Inverter electronic current check V |
| 6      | GND      | -   | VDD,VCC,VSS power Ground            |
| 7      | GND      | -   | VDD,VCC,VSS power Ground            |

### 5) CN5

| Pin no. | Pin name | I/O | Function                                  |
|---------|----------|-----|-------------------------------------------|
| 1       | DC+      | I   | Inverter electronic DC bus voltage check+ |
| 2       | NC       | -   |                                           |
| 3       | DC-      | I   | Inverter electronic DC bus voltage check- |

### 6) CN6

| Pin no | Pin name | I/O | Function                           |
|--------|----------|-----|------------------------------------|
| 1      | IMG1     | O   | Inverter electronic Magnet Control |
| 2      | IMG2     | O   | Inverter electronic Magnet Control |
| 3      | XOH      | I   | Inverter electronic radiant flat   |
| 4      | SCOM     | -   | Switch input power Ground          |

### 7) CN7 (OPP-2000 connector)

| Pin no | Pin name | I/O | Function                     |
|--------|----------|-----|------------------------------|
| 1      | GND      | -   | VDD power Ground             |
| 2      | TXD2     | O   | Data transmission of message |
| 3      | RXD2     | I   | Data receipt of message      |
| 4      | VDD      | -   | Digital power 5V             |

### 8) CN10 (GROUP & CRT communication connector)

| Pin no | Pin name | I/O | Function                        |
|--------|----------|-----|---------------------------------|
| 1      | CR-      | -   | CRT communication (RS485) DC-   |
| 2      | CR+      | -   | CRT communication (RS485) DC+   |
| 3      | DG-      | -   | Group communication (RS485) DG- |
| 4      | DG+      | -   | Group communication (RS485) DG+ |

### 9) CN11

| Pin no | Pin name | LED    | I/O | Function                                                           |
|--------|----------|--------|-----|--------------------------------------------------------------------|
| 1      | ICUT     | ICUT   | O   | Inverter power supply Isolated signal                              |
| 2      | DCOM     | DCOM   | O   | Door common Signal Output+ -                                       |
| 3      | BR1      | BR1    | O   | Output the control signal of Assistant Brake                       |
| 4      | BRCUT    | BRCUT  | O   | Output the forcing block signal of Brake                           |
| 5      | EQRST    | EQRST  | O   | EQ Sensor Reset Signal Output (When FPM7 is "O")                   |
|        | MPORT    |        |     | Multi function Output port (Select Output function from FPM7)      |
| 6      | DSOL     | DSOL   | O   | Output the Door Solenoid control                                   |
| 7      | MAGNET   | MAGNET | O   | Magnet control Output (In case Magnet control option is [YES] )    |
|        | RBRAKE   |        |     | Rope Brake Control Output (In case Magnet control option is [NO] ) |
| 8      | DCOMR    | DCOMR  | O   | Output the forcing common of door during using the Retreating CAM  |
| 9      | BRAKE    | BK     | O   | Relay Contact for Brake control Output                             |
| 10     | FAN      | FAN    | O   | Relay Contact for Fan Control Output                               |
| 11     | NC       | -      | -   |                                                                    |
| 12     | COM      | -      | -   | No.1 ~ 10 output commonness                                        |
| 13     | MHUSL    | MHUSL  | I   | Input the 5run UP limit switch                                     |

|    |                  |       |   |                                          |
|----|------------------|-------|---|------------------------------------------|
| 14 | VHDSL            | VHDSL | I | Input the 4run DOWN limit switch         |
| 15 | MHDSL            | MHDSL | I | Input the 5run DOWN limit switch         |
| 16 | PKS              | PKS   | I | Input the Parking switch                 |
| 17 | FIREMAN (RECALL) | FIRE2 | I | Input the emergency call switch (BS 2nd) |
| 18 | E-STOP           | ESTOP | I | Input the emergency stop switch          |
| 19 | RSV1             | FRS   | I | Not Specified                            |
| 20 | BRAKEM           | BKM   | I | Brake Monitoring                         |
| 21 | RSV2             | EINS  | I | Not Specified                            |
| 22 | INS              | INS   | I | Switch input for Auto/Manual change      |
| 23 | IUP              | IUP   | I | Switch input for manual UP driving       |
| 24 | IDN              | IDN   | I | Switch input for manual Down driving     |
| 25 | MGR(VH USL)      | VHUSL | I | Input the 4run UP limit switch           |

(Note) Output is MOSFET Relay and the maximum contact capacity is 100mA.

| Pin no | Pin name         | LED   | I/O | Function                             |
|--------|------------------|-------|-----|--------------------------------------|
| 26     | RBER             | RBER  | I   | Improper Input for Rope Brake        |
|        | FRS              |       |     | When FPM0<1, Fire Reset switch Input |
| 27     | DOOR             | DOOR  | I   | Switch input for Hall + Car door     |
| 28     | Normal Door Type | MK1K2 | I   | Input for Magnet Monitoring output   |
|        | Dual Door Type   |       | I   | B Door Open Limit Switch Input       |
| 29     | SCOM             | -     | -   | Switch common                        |
| 30     | SCOM             | -     | -   | Switch common                        |

### 10) CN13

| Pin no | Pin name | I/O | Function            |
|--------|----------|-----|---------------------|
| 1      | H1       | -   | IND,OPB Power AC18V |
| 2      | H20      | -   | IND,OPB Power AC18V |
| 3      | H1       | -   | IND,OPB Power AC18V |
| 4      | H20      | -   | IND,OPB Power AC18V |



### 11) CN14

| Pin no | Pin name | I/O | Function                    |
|--------|----------|-----|-----------------------------|
| 1      | C+12     | -   | IND,OPB Communication Power |
| 2      | NC       | -   |                             |
| 3      | C-12     | -   | IND,OPB Communication Power |

### 12) CN21

| Pin no | Pin name | I/O | Function                   |
|--------|----------|-----|----------------------------|
| 1      | G5V      | -   | IPM Gate Board Power DC5V  |
| 2      | H20      | -   | IGBT Control signal        |
| 3      | H1       | -   | VDD, VCC, VSS Power Ground |

### 13) J1

| Pin no | Pin name | I/O | Function                                      |
|--------|----------|-----|-----------------------------------------------|
| 1      | UPS      | I   | Input the UPS driving (power-cut controlling) |
| 2      | SCOM     | I   | Switch common                                 |
| 3      | SVL      | I   | Earthquake low input(earthquake)              |
| 4      | SL       | I   | Earthquake high input(earthquake)             |
| 5      | SCOM     | I   | Switch common                                 |
| 6      | SCOM     | I   | Switch common                                 |

### 14) CFO1

| Pin no | Pin name      | I/O | Function                          |
|--------|---------------|-----|-----------------------------------|
| 1      | IUS           | I   | Switch input for Inductor UP      |
| 2      | IDS           | I   | Switch input for Inductor DOWN    |
| 3      | DZONE         | I   | Switch input for Door Zone        |
| 4      | OP1 (OPEN)    | O   | Output the Door OPEN signal       |
| 5      | CL1 (CLOSE)   | O   | Output the Door CLOSE signal      |
| 6      | ACT1 (ACTIVE) | O   | ON during driving                 |
| 7      | COM1          | -   | OP1,CL1,ACT1 output commonness    |
| 8      | DOL           | I   | Switch input for Door Open Limit  |
| 9      | DCL           | I   | Switch input for Door Close Limit |
| 10     | SCOM          | I   | Switch common                     |
| 11     | SCOM          | I   | Switch common                     |
| 12     | SCOM          | I   | Switch common                     |

### 15) CFO3

| Pin no | Pin name | I/O | Function                    |
|--------|----------|-----|-----------------------------|
| 1      | H1       | -   | IND,OPB Power AC18V         |
| 2      | H1       | -   | IND,OPB Power AC18V         |
| 3      | H20      | -   | IND,OPB Power AC18V         |
| 4      | H20      | -   | IND,OPB Power AC18V         |
| 5      | C+12     | -   | IND,OPB Communication Power |
| 6      | C-12     | -   | IND,OPB Communication Power |
| 7      | RXD      | I   | IND,OPB Data Receive        |
| 8      | TXD      | O   | IND,OPB Data Send           |
| 9      | NC       | -   |                             |

### 16) CFO5

| Pin no | Pin name         |                   | I/O | Function                              |
|--------|------------------|-------------------|-----|---------------------------------------|
| 1      | Normal Door Type | CL2               | O   | Normal Close Signal Output            |
|        | Dual Door Type   | (OP2)<br>OPENr    |     | B Door Open Output                    |
| 2      | Normal Door Type | CL3               | O   | Handicapped Close Output              |
|        | Dual Door Type   | (CL2)<br>CLOSEr   |     | B Door Close Output                   |
| 3      | Normal Door Type | CL4               | O   | Output the Door CLOSE signal          |
|        | Dual Door Type   | (ACT2)<br>ACTIVER |     | ON during driving                     |
| 4      | COM2             |                   | -   | CL2,CL3,CL4<br>output commonness      |
| 5      | Normal Door Type | MK1K2             | I   | Input for Magnet<br>Monitoring output |
|        | Dual Door Type   | DOLr              |     | B Door Open Limit Switch Input        |
| 6      | Normal Door Type | ---               | I   | Not Specified                         |
|        | Dual Door Type   | DCLr              |     | B Door Close Limit Switch Input       |

### 17) CFH1

| Pin no | Pin name            | LED   | I/O | Function                                          |
|--------|---------------------|-------|-----|---------------------------------------------------|
| 1      | UL                  | UL    | I   | Input the Up limit switch                         |
| 2      | USL                 | USL   | I   | Input the 2run UP limit switch                    |
| 3      | HUSL                | HUSL  | I   | Input the 3run UP limit switch                    |
| 4      | VHUSL               | VHUSL | I   | Input the 4run UP limit switch                    |
| 5      | MHUSL               | MHUSL | I   | Input the 5run UP limit switch                    |
| 6      | DL                  | DL    | I   | Input the Down limit switch                       |
| 7      | DSL                 | DSL   | I   | Input the 2run Down limit switch                  |
| 8      | HDSL                | HDSL  | I   | Input the 3run Down limit switch                  |
| 9      | VHDSL               | VHDSL | I   | Input the 4run Down limit switch                  |
| 10     | MHDSL               | MHDSL | I   | Input the 5run Down limit switch                  |
| 11     | PKS                 | PKS   | I   | Input the Parking switch                          |
| 12     | FIREMAN<br>(RECALL) | FIRE2 | I   | Input the emergency call switch<br>(BS 2nd)       |
| 13     | FIRE                | FIRE1 | I   | Input the driving of fire controlling<br>(BS 1st) |
| 14     | SCOM                | -     | I   | Switch common                                     |
| 15     | SCOM                | -     | I   | Switch common                                     |

### 18) CFH3

| Pin no | Pin name | I/O | Function                    |
|--------|----------|-----|-----------------------------|
| 1      | H1       | -   | IND,OPB Power AC18V         |
| 2      | H1       | -   | IND,OPB Power AC18V         |
| 3      | H20      | -   | IND,OPB Power AC18V         |
| 4      | H20      | -   | IND,OPB Power AC18V         |
| 5      | C+12     | -   | IND,OPB Communication Power |
| 6      | C-12     | -   | IND,OPB Communication Power |
| 7      | RXD      | I   | IND,OPB Data Receive        |
| 8      | TXD      | O   | IND,OPB Data Send           |
| 9      | NC       | -   |                             |

### 19) J4

Tool contact connector for programming the DSP Chip

|                                                                                   |                            |          |                  |
|-----------------------------------------------------------------------------------|----------------------------|----------|------------------|
|  | <b>MMR Elevator</b>        | Rev. No. | 0                |
|                                                                                   |                            | Doc. No. | DL<br>EL) 06-007 |
| <b>FOD</b><br>Field Industrial Engineering                                        | <b>Installation Manual</b> | Page     | 92/107           |
|                                                                                   |                            | Date     | June. 2006       |

20) J3,J5,J6

Tool contact connector for programming the CPLD Chip

21) TB1

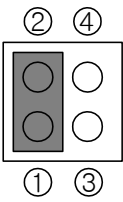
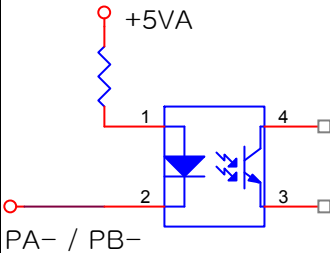
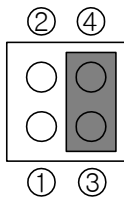
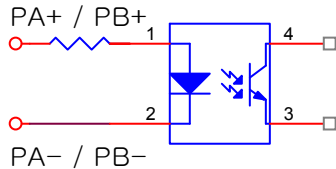
Used synchronous motor Encoder

22) TB2 (Encoder connection)

| Pin no | Pin name | I/O | Function                 |
|--------|----------|-----|--------------------------|
| 1      | GND      | -   | VCC power Ground         |
| 2      | VTFF     | I   | Useless terminal         |
| 3      | PA+      | I   | Encoder A phase +        |
| 4      | PA-      | I   | Encoder A phase -        |
| 5      | PB+      | I   | Encoder B phase +        |
| 6      | PB-      | I   | Encoder B phase -        |
| 7      | +5VA     | -   | Encoder power +5V        |
| 8      | SCOM     | -   | +5VA, +15VA power Ground |
| 9      | +12VA    | -   | Encoder power +12V       |
| 10     | SCOM     | -   | +5VA, +12VA power Ground |

### 3. Jumper Pin Explanation

- 1) JP1: Not use. Ship it out with Open condition:
- 2) JP2: In case of DSP program is short, and ship it out with open condition
- 3) JP3 & JP4 (Choose the encoder input type)

| JP3 & JP4                                                                           | Circuit                                                                             | JP3 & JP4                                                                           | Circuit                                                                               |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  |  |  |  |

### 4. DIP Switch Explanation

#### 1) Dip SW1

| SW1                                                                                                                                                                                         | ①   | ②   | ③        | ④   | Explanation                    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|----------|-----|--------------------------------|
| <div> <div>ON</div> <div>OFF</div> <div> <div>□</div> <div>□</div> <div>□</div> <div>□</div> </div> <div> <div>■</div> <div>■</div> <div>■</div> <div>■</div> </div> <div>①②③④</div> </div> | OFF | OFF | OFF      | -   | 1floor operation               |
|                                                                                                                                                                                             | OFF | OFF | ON       | -   | 2floors operation(60m/min)     |
|                                                                                                                                                                                             | OFF | ON  | OFF      | -   | 3floors operation(90~105m/min) |
|                                                                                                                                                                                             | OFF | ON  | (1)1. ON | -   | 4floors operation(120m/min)    |
|                                                                                                                                                                                             | ON  | OFF | OFF      | -   | 5floors operation(150m/min)    |
|                                                                                                                                                                                             | -   | -   | -        | OFF | B contact limit switch         |
|                                                                                                                                                                                             | -   | -   | -        | ON  | A contact limit switch         |

#### 2) Dip SW2

| SW1                                                                                                                                                                                         | ①   | ②   | ③   | ④   | Explanation                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|----------------------------------|
| <div> <div>ON</div> <div>OFF</div> <div> <div>□</div> <div>□</div> <div>□</div> <div>□</div> </div> <div> <div>■</div> <div>■</div> <div>■</div> <div>■</div> </div> <div>①②③④</div> </div> | OFF | -   | -   | -   | Simplex(SKA)                     |
|                                                                                                                                                                                             | ON  | -   | -   | -   | Duplex, Group(DSKA,N-PLEX)       |
|                                                                                                                                                                                             | -   | OFF | -   | -   | JIS fireman                      |
|                                                                                                                                                                                             | -   | ON  | -   | -   | BS South East Asia fireman       |
|                                                                                                                                                                                             | -   | -   | OFF | -   | ATT operation 1 (ATT1) – (note)1 |
|                                                                                                                                                                                             | -   | -   | ON  | -   | ATT operation 2 – (note)         |
|                                                                                                                                                                                             | -   | -   | -   | OFF | NP,EP,MGR,RTN 'No'               |
|                                                                                                                                                                                             | -   | -   | -   | ON  | NP,EP,MGR,RTN 'Yes'              |

note 1. ATT operation 1 (ATT1) - ATT,AUP,ADOWN Switch

note 2. ATT operation 2 (ATT2) - ATT,AUP,ADOWN,ASTART,APASS,ACUT Switch

|                                                                                   |                            |          |                  |
|-----------------------------------------------------------------------------------|----------------------------|----------|------------------|
|  | <b>MMR Elevator</b>        | Rev. No. | 0                |
|                                                                                   |                            | Doc. No. | DL<br>EL) 06-007 |
| <b>FOD</b><br>Field Industrial Engineering                                        | <b>Installation Manual</b> | Page     | 94/107           |
|                                                                                   |                            | Date     | June. 2006       |

### 5. LED Explanation

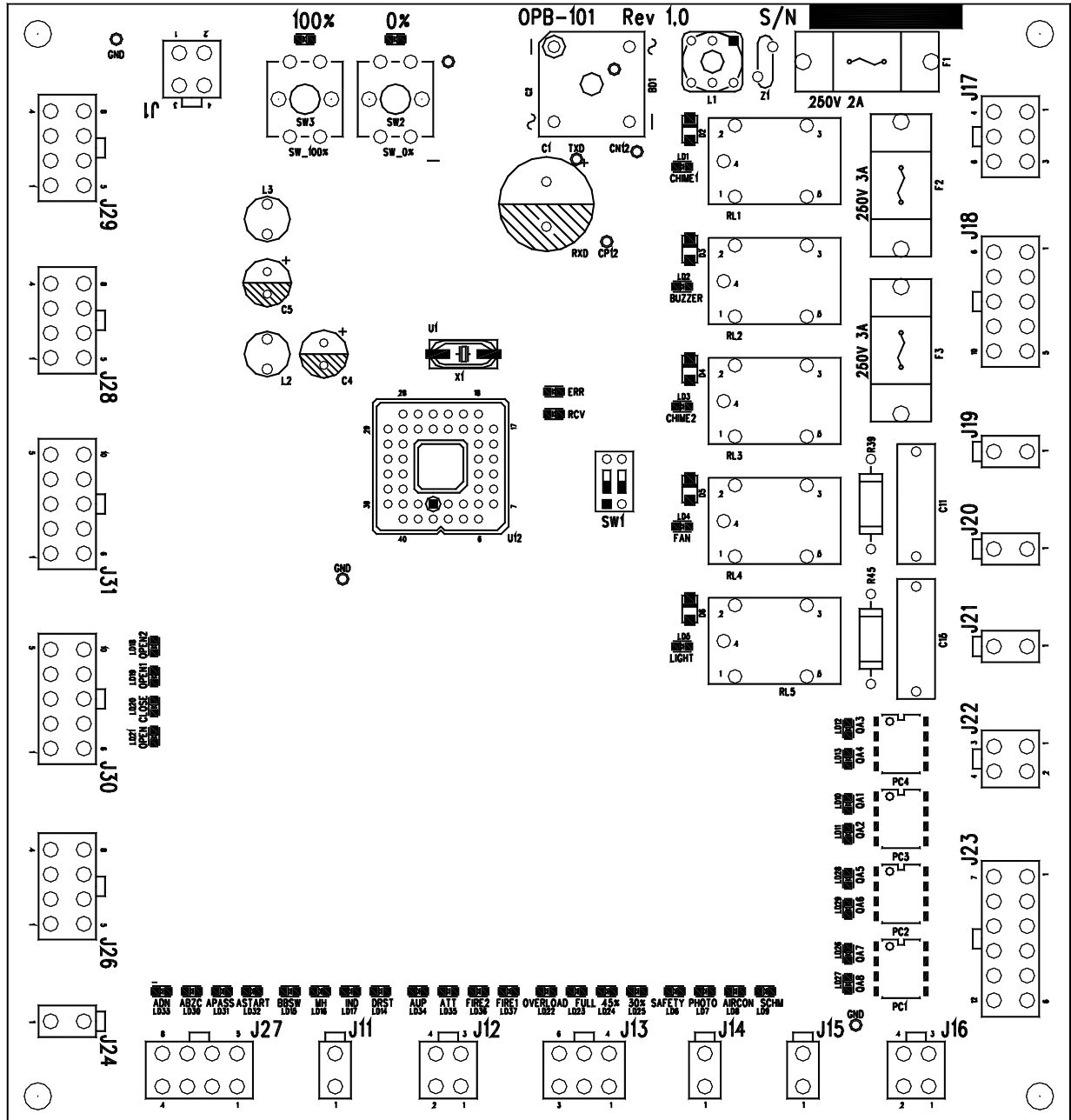
- 1) LD1: Be careful because of high voltage in Electrical pouts. When change LED for Inverter condenser is ON.
- 2) LD2: This is ON when Power is supplied to Digital Power condition display.
- 3) CTX: This is flickering when transmitting for CRT communication and is OFF when NO transmitting Data.
- 4) CRX: This is flickering when Receiving for CRT communication and is OFF when no receiving Data or Error.

### 6. Test Pin Explanation

- 1) GND: Digital Power 5V Ground.
- 2) CGND: IND, OPB Communication Power Ground.

### 3. 4. OPB-101 Board (Car top Interface Board)

#### 1. OPB-101 (Configuration)



### 2. Connector

#### 1) J11 (DOOR FORCED STOP SWITCH)

| Pin No | Pin Name | LED  | I/O | Function                      |
|--------|----------|------|-----|-------------------------------|
| 1      | DSTP     | LD14 | I   | Door Forced Stop Switch Input |
| 2      | GND      |      | -   | Switch Input Power Ground     |

#### 2) J12 (Fireman Switch Input)

| Pin No | Pin Name | LED  | I/O | Function               |
|--------|----------|------|-----|------------------------|
| 1      | FIRE1    | LD37 | I   | Fire 1 Input           |
| 2      | FIRE2    | LD36 | I   | Fire 2 Input           |
| 3      | +24V     | -    | -   | S/W Input Power 24V    |
| 4      | GND      | -    | -   | S/W Input Power Ground |

#### 3) J13 (LOAD S/W Input)

| Pin No | Pin Name | LED  | I/O | Function               |
|--------|----------|------|-----|------------------------|
| 1      | 30%      | LD25 | I   | Load 30% S/W Input     |
| 2      | 45%      | LD24 | I   | Load 45% S/W Input     |
| 3      | FULL     | LD23 | I   | FULL S/W Input         |
| 4      | OVERLOAD | LD22 | I   | OVERLOAD S/W Input     |
| 5      | NC       |      |     | No Used                |
| 6      | GND      | -    | -   | S/W Input Power Ground |

#### 4) J14 (SAFETY S/W Input)

| Pin No | Pin Name | LED | I/O | Function               |
|--------|----------|-----|-----|------------------------|
| 1      | SAFETY   | LD6 | I   | SAFETY S/W Input       |
| 2      | GND      |     | -   | S/W Input Power Ground |

#### 5) J15 (PHOTO S/W Input)

| Pin No | Pin Name | LED | I/O | Function               |
|--------|----------|-----|-----|------------------------|
| 1      | PHOTO    | LD7 | I   | PHOTO S/W Input        |
| 2      | GND      |     | -   | S/W Input Power Ground |



### 6) J16 (Relay Output)

| Pin No | Pin Name                                                                                 | LED  | I/O | Function                                             |
|--------|------------------------------------------------------------------------------------------|------|-----|------------------------------------------------------|
| 1      |  1<br>2 | LD26 | O   | When recall S/W is on, this contact becomes on.      |
| 2      |                                                                                          |      |     |                                                      |
| 3      |  3<br>4 | LD27 | O   | (JIS) When Fire1 S/W is on, this contact becomes on. |
| 4      |                                                                                          |      |     |                                                      |

\* . As MOSFET Relay Output, Contact Capacity is Maximum 100mA.

### 7) J17 (Power and Communication)

| Pin No | Pin Name | I/O | Function                 |
|--------|----------|-----|--------------------------|
| 1      | AC 0V    | -   | Board Input Power        |
| 2      | AC 18V   | -   | Board Input Power        |
| 3      | P24V     | -   | Communication Power +24V |
| 4      | N24V     | -   | Communication Power GND  |
| 5      | RXD      | I   | Data Receive             |
| 6      | TXD      | O   | Data Send                |

### 8) J18 (Control Output)

| Pin No | Pin Name             | LED | I/O | Function                               |
|--------|----------------------|-----|-----|----------------------------------------|
| 1      | DC24V+               |     | -   | Chime And Buzzer Power 24V+            |
| 2      | DC24V-               |     | -   | Chime And Buzzer Power 24V-            |
| 3      | Buzzer Power-        | LD2 | O   | Connect No3 and No4 with DC24V Buzzer. |
| 4      | Buzzer Power+        |     |     |                                        |
| 5      | DC24V+               |     | -   | DC24+ comes out through F2 Fuse.       |
| 6      | Chime Common Contact |     |     | Chime1 And Chime2 Common Terminal      |
| 7      | DC24V-               |     | -   | Chime And Buzzer Power 24V-            |
| 8      | Chime1 (UP)          | LD1 | O   | Up Chime Control Contact Output        |
| 9      | Chime2 (DOWN)        | LD3 | O   | Down Chime Control Contact Output      |
| 10     | NC                   |     |     | No Connection                          |

### 9) J19 (FAN And LIGHT Power)

| Pin No | Pin Name | I/O | Function                   |
|--------|----------|-----|----------------------------|
| 1      | AC 220V  | -   | Fan And Light Power AC220V |
| 2      | AC 0V    | -   | Fan And Light Power AC 0V  |

### 10) J20 (FAN Output)

| Pin No | Pin Name           | LED | I/O | Function                                |
|--------|--------------------|-----|-----|-----------------------------------------|
| 1      | FAN Control Output | LD4 | -   | Connect No.1 and No.2 with AC 220V Fan. |
| 2      |                    |     |     |                                         |

### 11) J21 (LIGHT Output)

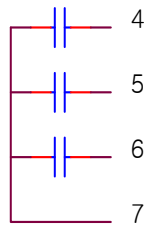
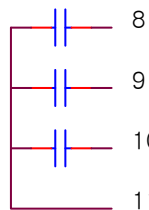
| Pin No | Pin Name             | LED | I/O | Function                                  |
|--------|----------------------|-----|-----|-------------------------------------------|
| 1      | LIGHT Control Output | LD5 | -   | Connect No.1 and No. 2 with AC220V Light. |
| 2      |                      |     |     |                                           |

### 12) J22 (FAN And LIGHT S/W)

| Pin No | Pin Name  | I/O | Function                                   |
|--------|-----------|-----|--------------------------------------------|
| 1      | FAN S/W   | -   | Connect No. 1 and No. 2 with FAN On/Off.   |
| 2      |           |     |                                            |
| 3      | LIGHT S/W | -   | Connect No. 1 and No. 2 with LIGHT On/Off. |
| 4      |           |     |                                            |

### 13) J23 (Relay Output)

| Pin No | Pin Name | LED  | I/O | Function                                        |
|--------|----------|------|-----|-------------------------------------------------|
| 1      | AIRCON   | LD8  | I   | AIRCON Sensor Input                             |
| 2      | SCHM     | LD9  | I   | Return Base floor S/W Input (More than PSD 4.3) |
| 3      | GND      | -    | -   | S/W Input Power Ground                          |
| 4      |          | LD10 | O   | Spare Output Contact 1                          |
| 5      |          | LD11 | O   | Spare Output Contact 2                          |
| 6      |          | LD12 | O   | Spare Output Contact 3                          |

|    |                                                                                   |      |   |                                                                                 |
|----|-----------------------------------------------------------------------------------|------|---|---------------------------------------------------------------------------------|
| 7  |  | -    | - | Spare Output Contact 1,2,3 Common                                               |
| 8  |  | LD13 | O | It becomes On for a second until door closes. (Close predict. More than PSD4.3) |
| 9  |                                                                                   | LD28 | O | When broken, buzzer output (More than PSD4.2)                                   |
| 10 |                                                                                   | LD29 | O | When a car stays on the lowest floor, this contact is on.                       |
| 11 |                                                                                   | -    | - | Spare output contact 4,5,6 com.                                                 |
| 12 | +24V                                                                              | -    | - | S/W Input Power 24V                                                             |

### 14) J24 (DC24V Output)

| Pin No | Pin Name | I/O | Function               |
|--------|----------|-----|------------------------|
| 1      | +24V     | -   | S/W Input Power 24V    |
| 2      | GND      | -   | S/W Input Power Ground |

### 15) J26 (Input)

| Pin No | Pin Name |                            | LED  | I/O | Function                                      |
|--------|----------|----------------------------|------|-----|-----------------------------------------------|
| 1      | ATT      |                            | LD35 | I   | Attendant Operation Selection S/W Input       |
| 2      | AUP      |                            | LD34 | I   | Attendant Up Operation Selection S/W Input    |
| 3      | ADOWN    |                            | LD33 | I   | Attendant Down Operation Selection S/W Input  |
| 4      | GND      |                            | -    | -   | S/W Input Power Ground                        |
| 5      | ASTART   |                            | LD32 | I   | Attendant Operation Start S/W Input           |
| 6      | APASS    | Dip SW2<br>③ is <b>On</b>  | LD31 | I   | When Attendant Operation, Hall Call Pass S/W  |
|        | ODD      | Dip SW2<br>③ is <b>Off</b> |      | I   | Odd floor Operation Selection Input(PSD 4.2)  |
| 7      | ABZC     | Dip SW2<br>③ is <b>On</b>  | LD30 | I   | When Attendant Operation, Buzzer Cut S/W      |
|        | EVEN     | Dip SW2<br>③ is <b>Off</b> |      | I   | Even floor Operation Selection Input(PSD 4.2) |
| 8      | GND      |                            | -    | -   | S/W Input Power Ground                        |

- J26 above means Dip S/W 2 of MCB-2001.

### 16) J27 (Input)

| Pin No | Pin Name | LED  | I/O | Function                                 |
|--------|----------|------|-----|------------------------------------------|
| 1      | IND      | LD17 | I   | Special Operation S/W Input              |
| 2      | MH       | LD16 | I   | Moving Operation S/W Input               |
| 3      | BBSW     | LD15 | I   | Each floor Operation Selection S/W Input |
| 4      | GND      |      | -   | S/W Input Power Ground                   |
| 5      | DSTP     | LD14 | I   | Door forced Stop S/W Input               |
| 6      | FIRE1    | LD37 | I   | Fire1 Operation S/W Input (JIS)          |
| 7      | FIRE2    | LD36 | I   | Fire1 Operation S/W Input (JIS)          |
| 8      | GND      |      | -   | S/W Power Ground                         |

### 17) J28, J29 (OPB-2000SP Connection)

| Pin No | Pin Name | I/O | Function               |
|--------|----------|-----|------------------------|
| 1      | +24V     | -   | S/W Input Power 24V    |
| 2      | +5V      | -   | Digital Power 5V       |
| 3      | GND      | -   | S/W Input Power Ground |
| 4      | DIN      | I   | Serial Data Input      |
| 5      | CLK      | O   | Serial Data Clock      |
| 6      | STRB     | O   | Serial Data Strobe     |
| 7      | DOUT     | O   | Serial Data Output     |
| 8      | GND      | -   | S/W Input Power Ground |

\* As J28 and J29 have the same function, it can be connector with any of them.

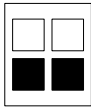
18) J30, J31 (Input)

| Pin No | Pin Name | LED  | I/O | Function                                                                            |
|--------|----------|------|-----|-------------------------------------------------------------------------------------|
| 1      | L41      | -    | O   | OPEN S/W Lamp Output                                                                |
| 2      | L42      | -    | O   | CLOSE S/W Lamp Output                                                               |
| 3      | L43      | -    | O   | When it has disable function, disable open S/W lamp output. Or Open S/W lamp output |
| 4      |          |      |     |                                                                                     |
| 5      | +24V     | -    | -   | S/W Input Power 24V                                                                 |
| 6      | OPEN     | LD21 | I   | OPEN S/W Input                                                                      |
| 7      | CLOSE    | LD20 | I   | CLOSE S/W Input                                                                     |
| 8      | OPEN1    | LD19 | I   | OPEN S/W Input                                                                      |
| 9      | OPEN2    | LD18 | I   | Disable OPEN S/W Input                                                              |
| 10     | GND      |      | -   | S/W Input Power Ground                                                              |

\* As J30 and J31 have the same function, it can be connector with any of them.

### 3. DIP S/W

1) Dip SW1

| SW1                                                                                                                                                                                      | ①   | ②   | Explanation                                                                             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----------------------------------------------------------------------------------------|
| ON <br>OFF <br>① ② | ON  | OFF | OPB-101 is selected for disable. So, every car call is functioning as disable car call. |
|                                                                                                                                                                                          | OFF | OFF | OPB-101 is selected for normal.                                                         |

\*Dip S/W No. 1 is used over 17 stories, because two pieces of OPB-101 should be used when disable function is included. And in that case S/W should be separated disable with normal.

\*Dip S/W No. 2 is not confined.

|                                                                                   |                            |          |                  |
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#### 4. LED

- 1) RCV: When receive is normal, it blinks and there's no receive, it turns off.
- 2) ERR: When receive is abnormal, it blinks and no receive, it turns on.

#### 5. Fuse

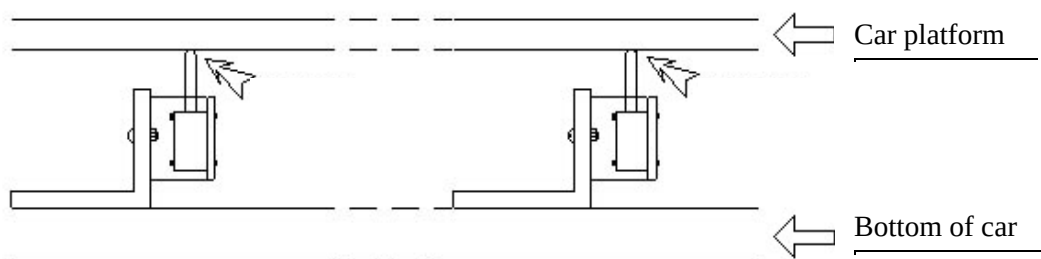
- 1) F1: Board Input Power Fuse about AC18V. (Input Power No1,2 of J17)
- 2) F2: Chime and Buzzer Power Fuse. (Input Power No1,2 of J18)
- 3) F3: Fan and Light Power Fuse. (Input Power No1,2 of J19)

#### 6. Load Sensor

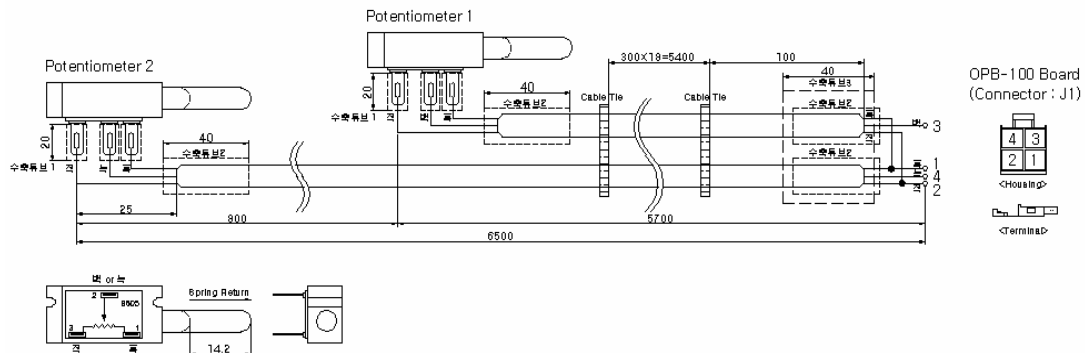
##### 6-1 Application:

##### 6-2. Installation of Analog Load Sensors

- 1) With empty car, install two analog load sensors between car platform and car bottom such that sensor bars nearly touch bottom of car platform.



2) Connect output signals from analog load sensors to J1 of OPB-101 REV1.

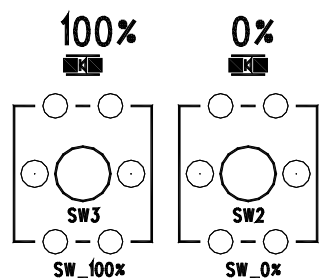


\* J1: (Analog Load Sensor CH0,CH1) input

| Pin | Pin name | I/O | Function              |
|-----|----------|-----|-----------------------|
| 1   | 0V       | -   | Load Sensor Power 0V  |
| 2   | +5V      | -   | Load Sensor Power +5V |
| 3   | CH0      | I   | Load Sensor CH0 Input |
| 4   | CH1      | I   | Load Sensor CH1 input |

### 6-3. Teaching of 0% and 100% load sensor value

- ▶ After teaching completed, inputted sensor value can be confirm using OPP-2000 (OPP3.0E) via "VIEW" -> 'SPM' -> 'SPM13'~'SPM18'( in case of MCB-2001Ci REV3.1)



- 1) With empty car, press 'SW-0%' button on OPB-101 REV1.0 board for more than 3 seconds. Then 'LD-0%' LED, which is initially on immediately after button press, flickers if sensor value is valid. When sensor value is

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invalid, LED which is initially on, turns off. With valid sensor value inputted, if LED flickers release button press to end 0% teaching.

- 2) With full car, press 'SW-100%' button on OPB-101 REV1.0 board for more than 3 seconds. Then 'LD-100%' LED, which is initially on immediately after button press, flickers if sensor value is valid. When sensor value is invalid, LED which is initially on, turns off. With valid sensor value inputted, if LED flickers release button press to end 100% teaching.

| Switch  | Status of LED                                          |                                                   | Remarks        |
|---------|--------------------------------------------------------|---------------------------------------------------|----------------|
|         | Valid sensor value                                     | Invalid sensor value                              |                |
| SW-100% | 'LD-100%' LED on<br>→ After 3 seconds,<br>LED flickers | 'LD-100%' LED on<br>→ After 3 seconds,<br>LED off | With full car  |
| SW-0%   | 'LD-0%' LED on<br>→ After 3 seconds,<br>LED flickers   | 'LD-0%' LED on<br>→ After 3 seconds,<br>LED off   | With empty car |

- Teaching of 0% and 100% load sensor value

6-4. Action when sensor value from Analog Load Sensor CH0 and CH1 is invalid

- 1) Check installation of load sensors, connection of output signals and load conditions are correct.

2) 0% teaching: Two sensor values must be less than 20% of full scale value of ADC which is 255. Thus if value of SPM17 and/or SPM18 is greater than or equal to 51, it's regarded as invalid conditions

- Action1: Check power connection (+ -) to sensors.
- Action2: Check whether sensor bars are abnormally too much pressed.

3) 100% teaching: If values of two sensors differ more than 30% of full scale value of ADC, which is 255, it's regarded as invalid conditions. Thus check whether values of SPM17 and SPM18 differ more than  $\pm 77$ .

- Action1: Check dummy loads are evenly distributed inside car.
- Action2: Check whether two sensor bars are equally pressed.



|                                                                                   |                            |          |                  |
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## 5. Appendix (Synchronous motor)

### 1. Lock System

Control panel of MMR must be once and for all locked(Synchronization of rotor of motor and encoder U, V, W signals) before any initial car movement. Lock should be done with no-load of motor which means ropes should not be on sheave of motor. To lock proceed as follows.

#### 1. Connection.

- Connect main power, R,S,T.
- Connect control panel(CP) and motor(M) as (CP, U – M, U), (CP, V – M, W), and (CP, W – M, V)
- Connect Earth.
- Connect brake signals, BRAKE COIL, BRAKE SWTCH to SEMR-100, J6.

|                                                                                   |                            |          |                  |
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- Connect ENCODER signals, PA+, PA-, PB+, PB-, U+, U-, V+, V-, W+, W-, +5V, N24.
- Connect safety lines and door lines according to appropriate regulations.

## 2. Setup after power-on.

- Connect IUS, IDS, LCM (CFO1/1, CFO1/2, CFO1/10).
- Software version of OPP-2000 should be above ver.3.1(Confirm via menu, 'OPP-2000 -> VER -> OPP-2000')
- Power on
- Set CPM parameters as follows
  - CPM 44: 60
  - CPM 45: 6
- After finishing other parameters which are necessary for proper operation, re-apply power

## 3. Commence Locking.

- Enter 'MODE' menu using OPP-2000.
- Select 'Lock'.
- When Locking begins magnet is energized, brake is open and motor sheave begins rotation to stop at "natural point" and then slowly rotates CW direction.
- If the operation, locking is successful, according to INSPECTION UP, DOWN button, sheave rotates appropriately
- Adjust CPM parameters, CPM 44, 45 according to parameter sheet

## 4. If Locking failed.

- Check encoder
  - Rotating sheave manually after brake open, check sign of SPM2 varies according to rotation direction. If it's o.k., then connection of A+, A-, B+, B- is correct. If not, check encoder itself or check connection
  - Looking in front of sheave, manually rotate motor sheave CW direction. During this operation, value of SPM9 should be changed as 32->96->64->192->128->160(1<sup>st</sup> value which is appear is not important. Sequence, however, is important). If it's o.k., U+, U-, V+, V-, W+, W- signal connection is correct. If not, check encoder itself or check connection
- After fixing problem, re-apply power and commence locking again.

|                                                                                   |                            |          |                  |
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- If lock still failed, interchange U, V connection to motor each other and, re-apply power and commence locking again

5. When car's UP and DOWN direction is reversed.

- Change value of CPM11, which is encoder direction change parameter and turn power off.
- With power off, interchange U, V connection to motor each other and, re-apply power and commence locking again.